

CLASSICAL VS PATH COUPLING ON A GEOMETRIC STATE SPACE: GLAUBER DYNAMICS FOR q -COLORINGS OF C_n

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ABSTRACT. We analyze one coupling of single-site Glauber dynamics for proper q -colorings of the cycle C_n in two ways. Drift on the Hamming disagreement count and path coupling on Hamming-adjacent pairs give the one-step contractions

$$\mathbb{E}[d_H(X_1, Y_1)] \leq \alpha_{\text{cl}}(q, n) d_H(X_0, Y_0), \quad \mathbb{E}[d_G(X_1, Y_1)] \leq \alpha_{\text{pc}}(q, n) d_G(X_0, Y_0),$$

with $\alpha_{\text{cl}}(q, n) = 1 - (q-4)/(n(q-2))$ and $\alpha_{\text{pc}}(q, n) = 1 - (q-6)/(n(q-2))$, both attained. The thresholds are $q \geq 5$ and $q \geq 7$, and the constants separate by $2/(n(q-2))$ uniformly in n . The gap is geometric. Transposing two colors across an edge produces a pair at Hamming distance 2 that no single recoloring connects, so the path metric d_G of the color graph charges a created disagreement +2 rather than +1; such pairs pack $\lfloor n/2 \rfloor$ to a cycle, giving $\text{diam } d_H = n$ and $\text{diam } d_G = \lfloor 3n/2 \rfloor$ for $q \geq 5$, hence closed $O(n \log n)$ bounds for both methods. The loss is attributable to the sparse edge set rather than to path coupling, since the complete edge set recovers α_{cl} . Exact enumeration on C_4 confirms every bound and gives worst-start $t_{\text{mix}}(0.05) = 57, 29, 22, 19, 18$ for $q = 3, \dots, 7$.

CONTENTS

1. Introduction	2
1.1. The chain	2
1.2. Results	3
2. Preliminaries	3
2.1. Markov chains	3
2.2. Total variation and mixing time	4
2.3. Couplings	5
2.4. Drift	6
2.5. Path coupling	6
2.6. Colorings and Glauber dynamics	7
2.7. The color graph	8
2.8. Enumeration by transfer matrix	8
3. Glauber dynamics on the four-cycle	9
3.1. The four-cycle and its state space	9
3.2. The chain and its ergodicity	10
3.3. Geometry of the four-cycle	10
4. Classical coupling bound	10
4.1. The synchronous maximal coupling	11
4.2. Sitewise matching	11
4.3. Contraction and mixing on the cycle	12
5. Path-coupling bound	13
5.1. Rigid pairs and the metric gap	13
5.2. Edge contraction in the path metric	14
6. Comparison, exact computation, and closed forms	15
6.1. Separation of the two bounds	15
6.2. The diameter of the color graph	16

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6.3. Exact computation on the four-cycle	18
6.4. Spectral analysis	19
6.5. Closed forms	20
Acknowledgments	20
Appendix A. Foundational tools	21
A.1. Linear algebra over $\mathbb{R}$	21
Appendix B. Reproducibility	21
B.1. Shared core	21
B.2. Figure 5	22
B.3. Figure 8	22
B.4. Figure 2	23
B.5. Verification	23
References	25

1. INTRODUCTION

Markov chain Monte Carlo underpins Bayesian statistics, statistical physics, and randomized approximation, and its guarantees are mixing-time bounds $t_{\text{mix}}(\varepsilon)$ certifying that the chain is ε -close to stationarity. Coupling is the standard route to such bounds [1, 2, 3, 5]. Path coupling [6] replaces the construction of a coalescing joint process by a one-step contraction verified only on adjacent pairs, and is the workhorse behind the coloring results of [10, 12, 13, 16]. The two methods are usually applied to different problems, so their relative strength is rarely measured. This paper fixes one chain, one coupling, and two analyses of it.

1.1. The chain. Fix $n \geq 3$, $q \geq 3$, and let $\Omega_q(C_n)$ be the proper q -colorings of the cycle. Identifying $V(C_n) = \mathbb{Z}/n$ and $[q] = \mathbb{Z}/q$, a coloring is a section of the discrete torus $\mathbb{Z}/n \times \mathbb{Z}/q$ whose consecutive values differ, that is, a closed loop winding once in the vertex direction and never running parallel to it. Figure 1 shows one for $n = 8$, $q = 5$. Counting these loops is a trace,

$$|\Omega_q(C_n)| = \text{tr}(T^n) = (q-1)^n + (-1)^n(q-1), \quad T_{cc'} = \mathbf{1}\{c \neq c'\},$$

the trace being exactly the gluing that closes a path into a loop (Theorem 2.8.1). Glauber dynamics moves one point of the loop at a time.

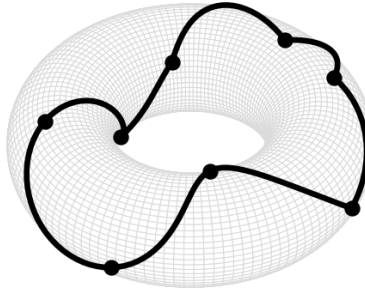


FIGURE 1. A proper coloring of C_8 with $q = 5$ as a closed loop on $\mathbb{Z}/8 \times \mathbb{Z}/5$.

1.2. Results. Throughout, (X_t, Y_t) is the synchronous maximal coupling of Definition 4.1.1, d_H the Hamming metric, and $d_{\mathcal{G}}$ the path metric of the color graph.

Theorem A. For all $n \geq 3$ and $q \geq 3$,

$$\mathbb{E}[d_H(X_1, Y_1)] \leq \alpha_{\text{cl}}(q, n) d_H(X_0, Y_0), \quad \alpha_{\text{cl}}(q, n) = 1 - \frac{q-4}{n(q-2)},$$

and $\alpha_{\text{cl}}(q, n) < 1$ if and only if $q \geq 5$.

Proof. Theorem 4.3.1, § 4. □

Theorem B. For all $n \geq 3$, $q \geq 4$, and every color-graph edge (X_0, Y_0) ,

$$\mathbb{E}[d_{\mathcal{G}}(X_1, Y_1)] \leq \alpha_{\text{pc}}(q, n) d_{\mathcal{G}}(X_0, Y_0), \quad \alpha_{\text{pc}}(q, n) = 1 - \frac{q-6}{n(q-2)},$$

and $\alpha_{\text{pc}}(q, n) < 1$ if and only if $q \geq 7$.

Proof. Theorem 5.2.1, § 5. □

Theorem C. For all $n \geq 3$ and $q \geq 4$,

$$\alpha_{\text{pc}}(q, n) - \alpha_{\text{cl}}(q, n) = \frac{2}{n(q-2)} > 0,$$

and on C_4 both constants are attained.

Proof. Corollary 5.2.4, § 5; Proposition 6.3.1, § 6. □

Theorem D. For all $n \geq 3$ and $q \geq 5$,

$$\text{diam } d_H = n, \quad \text{diam } d_{\mathcal{G}} = \left\lfloor \frac{3n}{2} \right\rfloor,$$

and consequently

$$\begin{aligned} q \geq 5 &\implies t_{\text{mix}}(\epsilon) \leq \left\lceil \frac{n(q-2)}{q-4} (\log(n) + \log(\epsilon^{-1})) \right\rceil, \\ q \geq 7 &\implies t_{\text{mix}}(\epsilon) \leq \left\lceil \frac{n(q-2)}{q-6} \left(\log\left(\left\lfloor \frac{3n}{2} \right\rfloor\right) + \log(\epsilon^{-1}) \right) \right\rceil. \end{aligned}$$

Proof. Theorem 6.2.3 and Theorem 6.5.1, § 6. □

2. PRELIMINARIES

Coupling for coloring chains goes back to Jerrum [10]; path coupling was introduced by Bubley and Dyer [6], and both are treated systematically in [5]. Everything used later is collected here, stated on C_n or a general graph G wherever the argument permits.

2.1. Markov chains.

Definition 2.1.1. Let P be a transition matrix on a finite set Ω . P is *irreducible* when every state reaches every other, *aperiodic* when $\gcd(\{t \geq 1 \mid P^t(x, x) > 0\}) = 1$ for every x , *stationary* for π when $\pi P = \pi$, and *reversible* for π when

$$\pi(x)P(x, y) = \pi(y)P(y, x) \quad \text{for all } x, y \in \Omega.$$

Theorem 2.1.2 (Convergence theorem). *If P is irreducible and aperiodic with stationary distribution π , then*

$$\lim_{t \rightarrow \infty} \max_{x \in \Omega} \|P^t(x, \cdot) - \pi\|_{\text{TV}} = 0.$$

Proof. [5, Theorem 4.9]. □

Lemma 2.1.3 (Reversibility implies stationarity). *If π is reversible for P , then $\pi P = \pi$.*

Proof.

$$(\pi P)(y) = \sum_{x \in \Omega} \pi(x) P(x, y) = \sum_{x \in \Omega} \pi(y) P(y, x) = \pi(y) \sum_{x \in \Omega} P(y, x) = \pi(y) \cdot 1 = \pi(y).$$

□

Corollary 2.1.4 (Symmetric chains). *If $P(x, y) = P(y, x)$ for all x, y , then the uniform measure is reversible, hence stationary.*

Proof.

$$\pi \equiv |\Omega|^{-1} \implies \pi(x) P(x, y) = \frac{P(x, y)}{|\Omega|} = \frac{P(y, x)}{|\Omega|} = \pi(y) P(y, x) \xrightarrow{2.1.3} \pi P = \pi.$$

□

2.2. Total variation and mixing time.

Definition 2.2.1. For probability measures μ, ν on Ω ,

$$\|\mu - \nu\|_{\text{TV}} = \frac{1}{2} \sum_{z \in \Omega} |\mu(z) - \nu(z)|.$$

Lemma 2.2.2 (Dual form).

$$\|\mu - \nu\|_{\text{TV}} = \max_{A \subseteq \Omega} (\mu(A) - \nu(A)).$$

Proof.

$$A \subseteq \Omega \implies \mu(A) - \nu(A) = \sum_{z \in A} (\mu(z) - \nu(z)) \leq \sum_{z \in A} (\mu(z) - \nu(z))^+ \leq \sum_{z \in \Omega} (\mu(z) - \nu(z))^+$$

$$A^* = \{z \mid \mu(z) > \nu(z)\} \implies \mu(A^*) - \nu(A^*) = \sum_{z \in \Omega} (\mu(z) - \nu(z))^+$$

$$\sum_z (\mu(z) - \nu(z)) = 1 - 1 = 0 \implies \sum_z (\mu(z) - \nu(z))^+ = \sum_z (\nu(z) - \mu(z))^+$$

$$|a - b| = (a - b)^+ + (b - a)^+ \implies \frac{1}{2} \sum_z |\mu(z) - \nu(z)| = \frac{1}{2} \left(2 \sum_z (\mu(z) - \nu(z))^+ \right) = \max_{A \subseteq \Omega} (\mu(A) - \nu(A)).$$

□

Lemma 2.2.3 (Overlap form).

$$\|\mu - \nu\|_{\text{TV}} = 1 - \sum_{z \in \Omega} \min\{\mu(z), \nu(z)\}.$$

Proof.

$$(\mu(z) - \nu(z))^+ = \max\{\mu(z), \nu(z)\} - \nu(z) = \mu(z) - \min\{\mu(z), \nu(z)\}$$

$$\xrightarrow{2.2.2} \|\mu - \nu\|_{\text{TV}} = \sum_{z \in \Omega} (\mu(z) - \min\{\mu(z), \nu(z)\}) = 1 - \sum_{z \in \Omega} \min\{\mu(z), \nu(z)\}.$$

□

Lemma 2.2.4 (Uniform measures). *For nonempty finite A, B with uniform measures U_A, U_B ,*

$$\|U_A - U_B\|_{\text{TV}} = 1 - |A \cap B| \min\{|A|^{-1}, |B|^{-1}\}.$$

Proof.

$$z \in A \cap B \implies \min\{U_A(z), U_B(z)\} = \min\{|A|^{-1}, |B|^{-1}\}, \quad z \notin A \cap B \implies \min\{U_A(z), U_B(z)\} = \min\{\cdot, 0\} = 0$$

$$\implies \sum_z \min\{U_A(z), U_B(z)\} = |A \cap B| \min\{|A|^{-1}, |B|^{-1}\} \xrightarrow{2.2.3} \|U_A - U_B\|_{\text{TV}} = 1 - |A \cap B| \min\{|A|^{-1}, |B|^{-1}\}.$$

□

Example 2.2.5. For $q = 5$, $A = \{1, 3, 4\}$, $B = \{1, 2\}$,

z	1	2	3	4	5
$U_A(z)$	$\frac{1}{3}$	0	$\frac{1}{3}$	$\frac{1}{3}$	0
$U_B(z)$	$\frac{1}{2}$	$\frac{1}{2}$	0	0	0
$\min\{U_A(z), U_B(z)\}$	$\frac{1}{3}$	0	0	0	0

$$\sum_{z \in [5]} \min\{U_A(z), U_B(z)\} = \frac{1}{3} + 0 + 0 + 0 + 0 = \frac{1}{3} \xrightarrow{2.2.3} \|U_A - U_B\|_{\text{TV}} = 1 - \frac{1}{3} = \frac{2}{3}$$

$$A \cap B = \{1\}, \quad |A \cap B| \min\{|A|^{-1}, |B|^{-1}\} = 1 \cdot \min\left\{\frac{1}{3}, \frac{1}{2}\right\} = \frac{1}{3} \xrightarrow{2.2.4} \|U_A - U_B\|_{\text{TV}} = 1 - \frac{1}{3} = \frac{2}{3}.$$

By Theorem 2.3.2 the overlap $\frac{1}{3}$ is $\mathbb{P}(X = Y)$ under the maximal coupling, so a mismatch occurs with probability $\frac{2}{3}$.

Definition 2.2.6.

$$d(t) = \max_{x \in \Omega} \|P^t(x, \cdot) - \pi\|_{\text{TV}}, \quad t_{\text{mix}}(\varepsilon) = \min\{t \mid d(t) \leq \varepsilon\}.$$

The numerical sections use the conventional value $\varepsilon = 0.05$.

2.3. Couplings. The coupling method originates with Doeblin [1] and was developed for mixing times by Aldous [2]; see [3, 5].

Definition 2.3.1. A *coupling* of μ, ν is a pair (X, Y) with $X \sim \mu$ and $Y \sim \nu$. A *coupling of the chain* is a process (X_t, Y_t) on Ω^2 whose coordinate marginals each evolve by P . It is *faithful* when $X_s = Y_s$ implies $X_t = Y_t$ for all $t \geq s$, with *coalescence time*

$$\tau = \min\{t \mid X_t = Y_t\}.$$

Theorem 2.3.2 (Maximal coupling). *Every μ, ν admit a coupling with*

$$\mathbb{P}(X \neq Y) = \|\mu - \nu\|_{\text{TV}},$$

and no coupling achieves less.

Proof. [5, Proposition 4.7]. □

Lemma 2.3.3 (Coupling inequality). *For a faithful coupling with $X_0 = x$ and $Y_0 \sim \pi$,*

$$\|P^t(x, \cdot) - \pi\|_{\text{TV}} \leq \mathbb{P}(\tau > t).$$

Proof.

$$X_t \sim P^t(x, \cdot), \quad Y_0 \sim \pi \xrightarrow{2.1.1} Y_t \sim \pi P^t = \pi$$

$$P^t(x, A) - \pi(A) = \mathbb{P}(X_t \in A) - \mathbb{P}(Y_t \in A) = \mathbb{P}(X_t \in A, Y_t \notin A) - \mathbb{P}(X_t \notin A, Y_t \in A)$$

$$\leq \mathbb{P}(X_t \in A, Y_t \notin A) \leq \mathbb{P}(X_t \neq Y_t) \xrightarrow{2.2.2} \|P^t(x, \cdot) - \pi\|_{\text{TV}} \leq \mathbb{P}(X_t \neq Y_t)$$

$$\{X_t \neq Y_t\} \xrightarrow{2.3.1} \{X_t \neq Y_t\} \subseteq \{\tau > t\} \implies \|P^t(x, \cdot) - \pi\|_{\text{TV}} \leq \mathbb{P}(\tau > t).$$

□

Remark 2.3.4. $\mathbb{E}[\tau]$ alone gives only $\mathbb{P}(\tau > t) \leq \mathbb{E}[\tau]/t$; the bounds below control $\mathbb{E}[D_t]$ at every t .

2.4. Drift.

Lemma 2.4.1 (Multiplicative drift). *Let (D_t) be adapted, nonnegative, integer valued, with $\mathbb{E}[D_{t+1} \mid \mathcal{F}_t] \leq \alpha D_t$ for some $\alpha \in (0, 1)$. Then*

$$\mathbb{E}[D_t] \leq \alpha^t D_0, \quad \mathbb{P}(D_t \geq 1) \leq \alpha^t D_0.$$

Proof.

$$\mathbb{E}[D_{t+1}] = \mathbb{E}[\mathbb{E}[D_{t+1} \mid \mathcal{F}_t]] \leq \mathbb{E}[\alpha D_t] = \alpha \mathbb{E}[D_t] \implies \mathbb{E}[D_t] \leq \alpha \mathbb{E}[D_{t-1}] \leq \alpha^2 \mathbb{E}[D_{t-2}] \leq \dots \leq \alpha^t D_0$$

$$\mathbb{P}(D_t \geq 1) = \sum_{k \geq 1} \mathbb{P}(D_t = k) \leq \sum_{k \geq 1} k \mathbb{P}(D_t = k) = \mathbb{E}[D_t] \leq \alpha^t D_0.$$

□

Corollary 2.4.2 (Drift to mixing). *If a faithful coupling satisfies the hypothesis of Lemma 2.4.1 with $\{D_t = 0\} = \{X_t = Y_t\}$ and $D_0 \leq D_{\max}$ for every starting pair, then*

$$d(t) \leq D_{\max} \alpha^t, \quad t_{\text{mix}}(\varepsilon) \leq \left\lceil \frac{\log D_{\max} + \log(\varepsilon^{-1})}{-\log(\alpha)} \right\rceil.$$

Proof.

$$\begin{aligned} \{\tau > t\} &= \{D_t \geq 1\} \xrightarrow{2.4.1} \mathbb{P}(\tau > t) \leq \alpha^t D_0 \leq D_{\max} \alpha^t \xrightarrow{2.3.3} d(t) \leq D_{\max} \alpha^t \\ D_{\max} \alpha^t \leq \varepsilon &\iff \log(D_{\max}) + t \log(\alpha) \leq \log(\varepsilon) \iff t \log(\alpha^{-1}) \geq \log(D_{\max}) + \log(\varepsilon^{-1}). \end{aligned}$$

□

2.5. Path coupling. Variants of Theorem 2.5.3 with longer paths and stopping times appear in [8, 9]. Proposition 2.5.2 and its corollaries are instantiated with $\rho = d_H$, $d_{\mathcal{E}} = d_{\mathcal{G}}$ in Lemma 2.7.2.

Definition 2.5.1. Let $\mathcal{E} \subseteq \Omega \times \Omega$ be symmetric with connected graph and weights $w \geq 1$. The *induced path metric* is

$$d_{\mathcal{E}}(x, y) = \min \left\{ \sum_{i=1}^m w(x_{i-1}, x_i) \mid x = x_0, \dots, x_m = y, (x_{i-1}, x_i) \in \mathcal{E} \right\}.$$

Proposition 2.5.2 (Maximality). *If a metric ρ satisfies $\rho \leq w$ on $\mathcal{E}$, then $\rho \leq d_{\mathcal{E}}$ pointwise.*

Proof.

$$x = x_0, \dots, x_m = y \text{ in } \mathcal{E} \implies \rho(x_0, x_m) \leq \rho(x_0, x_1) + \rho(x_1, x_m) \leq \rho(x_0, x_1) + \rho(x_1, x_2) + \rho(x_2, x_m) \leq \dots \leq \sum_{i=1}^m \rho(x_{i-1}, x_i)$$

$$\leq \sum_{i=1}^m w(x_{i-1}, x_i) \xrightarrow{2.5.1} \rho(x, y) \leq d_{\mathcal{E}}(x, y).$$

□

Theorem 2.5.3 (Bubley–Dyer). *Let $d = d_{\mathcal{E}}$, and suppose every $(x, y) \in \mathcal{E}$ admits a one-step coupling with $\mathbb{E}[d(X_1, Y_1)] \leq \alpha d(x, y)$ for some $\alpha < 1$. Then the contraction holds for all pairs, and with $d_{\max} = \max_{x, y} d(x, y)$,*

$$t_{\text{mix}}(\varepsilon) \leq \left\lceil \frac{\log d_{\max} + \log(\varepsilon^{-1})}{-\log(\alpha)} \right\rceil.$$

Proof. [6]; see also [5, Theorem 14.6].

□

Corollary 2.5.4 (Edge-only verification is never sharper). *Fix a one-step coupling and $w \equiv 1$. For any metric ρ with $\rho = 1$ on $\mathcal{E}$, put $\alpha_{\rho} = \max_{\mathcal{E}}(\mathbb{E}[\rho(X_1, Y_1)])$ and $\alpha_{\mathcal{E}} = \max_{\mathcal{E}}(\mathbb{E}[d_{\mathcal{E}}(X_1, Y_1)])$. Then $\alpha_{\rho} \leq \alpha_{\mathcal{E}}$.*

Proof.

$$\rho = 1 = w \text{ on } \mathcal{E} \xrightarrow{2.5.2} \rho(z, z') \leq d_{\mathcal{E}}(z, z') \quad \forall z, z' \implies \mathbb{E}[\rho(X_1, Y_1)] \leq \mathbb{E}[d_{\mathcal{E}}(X_1, Y_1)] \text{ on } \mathcal{E} \implies \alpha_{\rho} \leq \alpha_{\mathcal{E}}.$$

□

Corollary 2.5.5 (Complete edge set recovers drift). *For a metric $\rho \geq 1$ off the diagonal, $\mathcal{E} = \{(x, y) \mid x \neq y\}$, and $w = \rho$,*

$$d_{\mathcal{E}} = \rho,$$

and Theorem 2.5.3 reduces to Corollary 2.4.2.

Proof.

$$\begin{aligned} (x, y) \in \mathcal{E} &\xrightarrow{2.5.1} d_{\mathcal{E}}(x, y) \leq w(x, y) = \rho(x, y), \quad \rho = w \text{ on } \mathcal{E} \xrightarrow{2.5.2} \rho(x, y) \leq d_{\mathcal{E}}(x, y) \\ &\implies d_{\mathcal{E}}(x, y) \leq \rho(x, y) \leq d_{\mathcal{E}}(x, y) \implies d_{\mathcal{E}} = \rho. \end{aligned}$$

□

Remark 2.5.6. The complete edge set reproduces the drift constant, so a sparse $\mathcal{E}$ costs exactly the passage $\rho \rightsquigarrow d_{\mathcal{E}}$; § 5 computes this gap on $\Omega_q(C_4)$.

2.6. Colorings and Glauber dynamics. Glauber dynamics for colorings enters the mixing literature with [10, 11]; the definitions follow [5, § 3.3].

Definition 2.6.1. Write $[q] = \{1, 2, \dots, q\}$. For a graph G with maximum degree Δ , a *proper q -coloring* is a map $x: V(G) \rightarrow [q]$ with $x(u) \neq x(v)$ whenever $u \sim v$, and $\Omega_q(G)$ denotes the set of these. The *legal set* at v is

$$L_x(v) = \{c \in [q] \mid c \neq x(u) \text{ for all } u \sim v\}.$$

Glauber dynamics selects $v \in V(G)$ uniformly and recolors v uniformly from $L_x(v)$.

Lemma 2.6.2 (Legal sets on cycles). *For $x \in \Omega_q(C_n)$ with $n \geq 3$ and any vertex v ,*

$$L_x(v) = [q] \setminus \{x(v-1), x(v+1)\}, \quad |L_x(v)| \in \{q-2, q-1\}, \quad x(v) \in L_x(v).$$

Proof.

$$\begin{aligned} N(v) = \{v-1, v+1\} &\xrightarrow{2.6.1} L_x(v) = [q] \setminus \{x(v-1), x(v+1)\} \\ |\{x(v-1), x(v+1)\}| \in \{1, 2\} &\implies |L_x(v)| = q - |\{x(v-1), x(v+1)\}| \in \{q-2, q-1\} \\ x \text{ proper} &\implies x(v) \neq x(v-1), \quad x(v) \neq x(v+1) \implies x(v) \notin \{x(v-1), x(v+1)\} \implies x(v) \in L_x(v). \end{aligned}$$

□

Lemma 2.6.3 (Symmetry of Glauber dynamics). *On $\Omega_q(G)$ with $|V(G)| = n$, the Glauber transition matrix satisfies $P(x, y) = P(y, x)$ for all x, y . The uniform measure is reversible and stationary, and the chain is aperiodic.*

Proof.

$$\begin{aligned} x = y \text{ off } v &\implies x|_{N(v)} = y|_{N(v)} \xrightarrow{2.6.1} L_x(v) = L_y(v) \\ \implies P(x, y) &= \mathbb{P}(\text{select } v) \cdot U_{L_x(v)}(y(v)) = \frac{1}{n} \cdot \frac{1}{|L_x(v)|} = \frac{1}{n} \cdot \frac{1}{|L_y(v)|} = P(y, x) \\ d_H(x, y) \geq 2 &\implies P(x, y) = 0 = P(y, x) \xrightarrow{2.1.4} \pi \text{ uniform reversible} \\ x(v) \in L_x(v) &\implies P(x, x) \geq \frac{1}{n} \cdot \frac{1}{|L_x(v)|} > 0 \implies \gcd(\{t \geq 1 \mid P^t(x, x) > 0\}) = 1. \end{aligned}$$

□

2.7. The color graph. The color graph is the reconfiguration graph of colorings [21, 23, 24, 22]; see [25, 26] for surveys. § 5 contracts its metric $d_{\mathcal{G}}$ and § 6.2 computes its diameter on cycles.

Definition 2.7.1. The *color graph* $\mathcal{C}_q(G)$ has vertex set $\Omega_q(G)$, with $x \sim y$ when $d_H(x, y) = 1$. Its graph metric is $d_{\mathcal{G}}$.

Lemma 2.7.2 (Glauber moves are color-graph edges). *For $x \neq y$, $P(x, y) > 0$ if and only if $(x, y) \in E(\mathcal{C}_q(G))$. In particular the chain is irreducible if and only if $\mathcal{C}_q(G)$ is connected, and*

$$d_H \leq d_{\mathcal{G}} \quad \text{pointwise.}$$

Proof.

$$\begin{aligned} P(x, y) > 0, x \neq y &\stackrel{2.6.1}{\iff} d_H(x, y) = 1, y(v) \in L_x(v) \stackrel{2.7.1}{\iff} (x, y) \in E(\mathcal{C}_q(G)) \\ d_H &= 1 \text{ on } E(\mathcal{C}_q(G)) \stackrel{2.5.2}{\implies} d_H \leq d_{\mathcal{G}}. \end{aligned}$$

□

Theorem 2.7.3 (Connectivity of the color graph). *If $q \geq \Delta + 2$, then $\mathcal{C}_q(G)$ is connected. In particular Glauber dynamics on $\Omega_q(C_n)$ is irreducible for $q \geq 4$.*

Proof. [21, Theorem 1]; distances in $\mathcal{C}_q(G)$ are studied in [22].

□

Remark 2.7.4. For $3 \mid n$, the coloring $(1, 2, 3, 1, 2, 3, \dots) \in \Omega_3(C_n)$ has $L_x(v) = \{x(v)\}$ at every v , hence is frozen and disconnects $\mathcal{C}_3(C_n)$ [23, 24]; $\mathcal{C}_3(C_4)$ is connected by enumeration of its 18 states, so $q \geq \Delta + 2$ is sharp on cycles.

2.8. Enumeration by transfer matrix. The formula is classical [27], by transfer matrix [28, Chapter 9]; the trace decomposition below is reused in Proposition 3.1.1 and Remark 3.1.2.

Theorem 2.8.1 (Chromatic polynomial of a cycle). *For $n \geq 3$, define $T \in \mathbb{Z}^{q \times q}$ by $T_{cc'} = \mathbf{1}\{c \neq c'\}$ for $c, c' \in [q]$. Then*

$$|\Omega_q(C_n)| = \text{tr}(T^n) = (q-1)^n + (-1)^n(q-1).$$

Proof.

$$\begin{aligned} q = 3 &\implies T = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}, \quad T^2 = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}, \\ T^3 &= \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 3 & 3 \\ 3 & 2 & 3 \\ 3 & 3 & 2 \end{bmatrix} \implies \text{tr}(T^3) = 2 + 2 + 2 = 6 = (3-1)^3 + (-1)^3(3-1) \\ \text{general } q, n : &\quad (T^n)_{c_0 c_n} = \sum_{c_1 \in [q]} \cdots \sum_{c_{n-1} \in [q]} T_{c_0 c_1} T_{c_1 c_2} \cdots T_{c_{n-1} c_n} = \#\{(c_1, \dots, c_{n-1}) \mid c_i \neq c_{i+1} \text{ for all } i\} \\ &\implies \text{tr}(T^n) = \sum_{c \in [q]} (T^n)_{cc} = \#\{(c_0, \dots, c_{n-1}) \in [q]^n \mid c_i \neq c_{i+1 \bmod n} \text{ for all } i\} = |\Omega_q(C_n)| \\ \mathbf{1} = (1, 1, \dots, 1)^\top &\implies (T\mathbf{1})_c = \sum_{c' \in [q]} T_{cc'} = \#\{c' \mid c' \neq c\} = q-1 \implies T\mathbf{1} = (q-1)\mathbf{1} \\ v \perp \mathbf{1} &\implies (Tv)_c = \sum_{c' \neq c} v_{c'} = \left(\sum_{c' \in [q]} v_{c'} \right) - v_c = 0 - v_c = -v_c \implies Tv = -v \\ \dim(\{v \mid v \perp \mathbf{1}\}) &= q-1 \implies \text{tr}(T^n) = (q-1)^n \cdot 1 + (-1)^n \cdot (q-1). \end{aligned}$$

□

Example 2.8.2.

$\text{tr}(T^3) = 6$, $\Omega_3(C_3) = \{(1, 2, 3), (1, 3, 2), (2, 1, 3), (2, 3, 1), (3, 1, 2), (3, 2, 1)\}$, $|\Omega_3(C_3)| = 3! = 6$,
the proper 3-colorings of a triangle being the bijections $[3] \rightarrow [3]$.

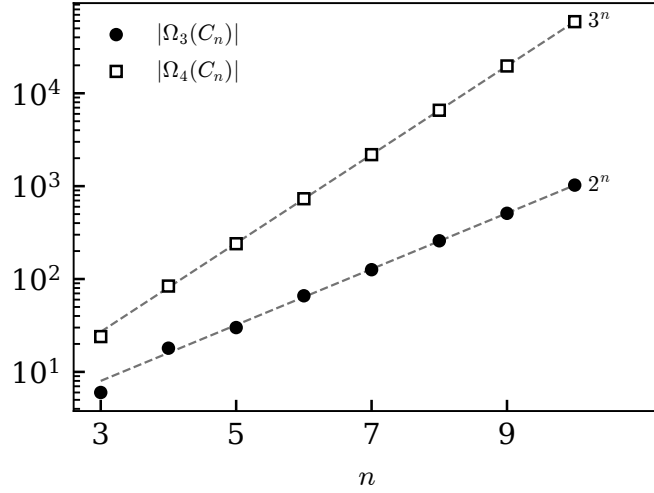


FIGURE 2. $|\Omega_q(C_n)| = (q-1)^n + (-1)^n(q-1)$ for $q=3$ (filled) and $q=4$ (open) on a logarithmic scale; dashed, the asymptotes $(q-1)^n$.

3. GLAUBER DYNAMICS ON THE FOUR-CYCLE

This section specializes § 2 to $G = C_4$.

3.1. The four-cycle and its state space. Let $V(C_4) = \{0, 1, 2, 3\}$ and $E(C_4) = \{(i, i+1 \bmod 4) \mid i \in V(C_4)\}$, all vertex arithmetic modulo 4. By Definition 2.6.1,

$$\Omega_q(C_4) = \{x \in [q]^{V(C_4)} \mid x(i) \neq x(i+1 \bmod 4) \ \forall i \in V(C_4)\}.$$

Proposition 3.1.1 (State-space size). *For $q \geq 2$, with $T \in \mathbb{Z}^{q \times q}$, $T_{cc'} = \mathbf{1}\{c \neq c'\}$, as in Theorem 2.8.1,*

$$|\Omega_q(C_4)| = q(q-1)^2 + q(q-1)(q-2)^2 = (q-1)^4 + (q-1) = q(q-1)(q^2 - 3q + 3).$$

Proof.

$$\begin{aligned} n=4 &\xrightarrow{2.8.1} |\Omega_q(C_4)| = \text{tr}(T^4) = (q-1)^4 + (-1)^4(q-1) = (q-1)^4 + (q-1) \\ (T^2)_{cc'} &= \sum_{b \in [q]} T_{cb}T_{bc'} = \#\{b \in [q] \mid b \neq c, b \neq c'\} = (q-2) + \mathbf{1}\{c = c'\} \\ T = T^\top &\implies (T^2)^\top = T^2 \implies \text{tr}(T^4) = \sum_{c \in [q]} \sum_{c' \in [q]} (T^2)_{cc'}(T^2)_{c'c} = \sum_{c \in [q]} \sum_{c' \in [q]} ((T^2)_{cc'})^2 \\ c = c' &\implies ((T^2)_{cc})^2 = (q-1)^2, \quad c \neq c' \implies ((T^2)_{cc'})^2 = (q-2)^2 \\ \implies \text{tr}(T^4) &= \sum_{c \in [q]} (q-1)^2 + \sum_{c \in [q]} \sum_{c' \in [q] \setminus \{c\}} (q-2)^2 = q(q-1)^2 + q(q-1)(q-2)^2 \\ q(q-1)^2 + q(q-1)(q-2)^2 &= q(q-1)((q-1) + (q-2)^2) = q(q-1)(q-1 + q^2 - 4q + 4) = q(q-1)(q^2 - 3q + 3). \end{aligned}$$

□

In particular $|\Omega_q(C_4)| = 18, 84, 260, 630, 1302$ for $q = 3, 4, 5, 6, 7$.

Remark 3.1.2. In the trace computation $(T^2)_{cc'}$ counts the choices of $x(1)$ given $x(0) = c, x(2) = c'$, so

$$|\Omega_q(C_4)| = \underbrace{q(q-1)^2}_{x(0)=x(2)} + \underbrace{q(q-1)(q-2)^2}_{x(0) \neq x(2)}.$$

The count $q(q-1)(q-2)^2$ is the off-diagonal term alone; by Lemma 2.6.2 the omitted colorings are exactly those possessing a legal set of size $q-1$.

3.2. The chain and its ergodicity. The chain of the paper is the Glauber dynamics of Definition 2.6.1 on $G = C_4$. By Lemma 2.6.2, for $x \neq y$ differing exactly at v ,

$$P(x, y) = \frac{1}{4|L_x(v)|} \mathbf{1}\{y(v) \in L_x(v)\}, \quad d_H(x, y) \geq 2 \implies P(x, y) = 0.$$

Corollary 3.2.1 (Ergodicity on the four-cycle). *For every $q \geq 3$, the Glauber chain on $\Omega_q(C_4)$ is aperiodic, irreducible, and reversible for the uniform measure $\pi \equiv |\Omega_q(C_4)|^{-1}$. Consequently*

$$\lim_{t \rightarrow \infty} d(t) = 0.$$

Proof.

$$\begin{aligned} q \geq 4 &\xrightarrow{2.7.3} d_{\mathcal{G}}(x, y) < \infty \quad \forall x, y \in \Omega_q(C_4), \quad q = 3 \xrightarrow{2.7.4} d_{\mathcal{G}}(x, y) < \infty \quad \forall x, y \in \Omega_3(C_4) \\ &\xrightarrow{2.7.2} \forall x, y \in \Omega_q(C_4) \quad \exists t \geq 0 \quad P^t(x, y) > 0 \\ x(v) \in L_x(v) &\xrightarrow{2.6.2} P(x, x) \geq \frac{1}{4|L_x(v)|} > 0 \implies \gcd(\{t \geq 1 \mid P^t(x, x) > 0\}) = 1 \\ &\xrightarrow{2.6.3} P(x, y) = P(y, x) \xrightarrow{2.1.4} \pi P = \pi \xrightarrow{2.1.2} \lim_{t \rightarrow \infty} d(t) = 0. \end{aligned}$$

□

3.3. Geometry of the four-cycle.

Lemma 3.3.1 (Opposite vertices share neighborhoods). *For every $v \in V(C_4)$,*

$$N(v) = \{v-1, v+1\} = N(v+2), \quad (v-1, v+1) \notin E(C_4).$$

Proof.

$$\begin{aligned} E(C_4) = \{(i, i+1 \bmod 4) \mid i \in V(C_4)\} &\implies N(v) = \{v-1, v+1\}, \quad N(v+2) = \{v+1, v+3\} = \{v+1, v-1\} \\ (v+1) - (v-1) = 2 &\not\equiv \pm 1 \pmod{4} \implies (v-1, v+1) \notin E(C_4). \end{aligned}$$

□

Corollary 3.3.2 (Shared legal sets across a single disagreement). *If $x, y \in \Omega_q(C_4)$ agree on $V(C_4) \setminus \{v\}$, then*

$$L_x(v) = L_y(v), \quad L_x(v+2) = L_y(v+2).$$

Proof.

$$\begin{aligned} N(v) = N(v+2) = \{v-1, v+1\} &\subseteq V(C_4) \setminus \{v\} \xrightarrow{3.3.1} x|_{N(v)} = y|_{N(v)}, \quad x|_{N(v+2)} = y|_{N(v+2)} \\ &\xrightarrow{2.6.1} L_x(v) = L_y(v), \quad L_x(v+2) = L_y(v+2). \end{aligned}$$

□

4. CLASSICAL COUPLING BOUND

Coupling goes back to Doeblin [1] and was developed for mixing times by Aldous [2]; see [3, 5], and [10] for colorings. This section builds one coupling on $\Omega_q(C_n)$ and analyzes it by drift on the Hamming disagreement count; § 5 reanalyzes the same coupling by path coupling.

4.1. The synchronous maximal coupling.

Definition 4.1.1 (Synchronous maximal coupling). Given $(X_0, Y_0) \in \Omega_q(C_n)^2$, one step draws $u \in V(C_n)$ uniformly, recolors u in both chains by a maximal coupling of $U_{L_{X_0}(u)}$ and $U_{L_{Y_0}(u)}$ as in Theorem 2.3.2, and leaves every other vertex unchanged. Write

$$S_t = \{v \in V(C_n) \mid X_t(v) \neq Y_t(v)\}, \quad D_t = |S_t| = d_H(X_t, Y_t), \quad T_u = \|U_{L_{X_0}(u)} - U_{L_{Y_0}(u)}\|_{TV}.$$

Lemma 4.1.2 (One-step identity). *Definition 4.1.1 yields a faithful coupling of the Glauber chain, and conditional on (X_0, Y_0) and the drawn vertex u ,*

$$D_1 = D_0 - \mathbf{1}\{u \in S_0\} + B_u, \quad B_u \sim \text{Bern}(T_u).$$

Proof.

$$\begin{aligned} X_1(u) &\sim U_{L_{X_0}(u)}, \quad Y_1(u) \sim U_{L_{Y_0}(u)} \xrightarrow{2.6.1} \text{each coordinate marginal evolves by } P \\ X_0 = Y_0 &\implies L_{X_0}(u) = L_{Y_0}(u) \xrightarrow{2.3.2} \mathbb{P}(X_1(u) \neq Y_1(u)) = \|U_{L_{X_0}(u)} - U_{L_{Y_0}(u)}\|_{TV} = 0 \implies X_1 = Y_1 \\ X_1|_{V \setminus \{u\}} &= X_0|_{V \setminus \{u\}}, \quad Y_1|_{V \setminus \{u\}} = Y_0|_{V \setminus \{u\}} \implies S_1 \setminus \{u\} = S_0 \setminus \{u\} \\ &\implies D_1 = |S_0 \setminus \{u\}| + \mathbf{1}\{X_1(u) \neq Y_1(u)\} = D_0 - \mathbf{1}\{u \in S_0\} + \mathbf{1}\{X_1(u) \neq Y_1(u)\} \\ &\xrightarrow{2.3.2} \mathbb{P}(X_1(u) \neq Y_1(u) \mid X_0, Y_0, u) = T_u \implies \mathbf{1}\{X_1(u) \neq Y_1(u)\} \sim \text{Bern}(T_u). \end{aligned}$$

□

4.2. Sitewise matching.

Lemma 4.2.1 (Matching bound). *For every $u \in V(C_n)$, with $r_u = |N(u) \cap S_0|$,*

$$T_u \leq \frac{r_u}{q-2}.$$

Proof.

$$\begin{aligned} F_X &= \{X_0(w) \mid w \in N(u)\}, \quad F_Y = \{Y_0(w) \mid w \in N(u)\}, \quad X_0 = Y_0 \text{ on } N(u) \setminus S_0 \\ &\implies |F_Y \setminus F_X| \leq r_u \implies |F_X \cup F_Y| = |F_X| + |F_Y \setminus F_X| \leq \min\{|F_X|, |F_Y|\} + r_u \\ L_{X_0}(u) &= [q] \setminus F_X, \quad L_{Y_0}(u) = [q] \setminus F_Y \implies |L_{X_0}(u) \cap L_{Y_0}(u)| = q - |F_X \cup F_Y| \geq q - \min\{|F_X|, |F_Y|\} - r_u \\ q - \min\{|F_X|, |F_Y|\} &= \max\{|L_{X_0}(u)|, |L_{Y_0}(u)|\} =: M \implies |L_{X_0}(u) \cap L_{Y_0}(u)| \geq M - r_u \\ \min\{|L_{X_0}(u)|^{-1}, |L_{Y_0}(u)|^{-1}\} &= M^{-1} \xrightarrow{2.2.4} T_u = 1 - \frac{|L_{X_0}(u) \cap L_{Y_0}(u)|}{M} \leq 1 - \frac{M - r_u}{M} = \frac{r_u}{M} \\ &\xrightarrow{2.6.2} M \geq q - 2 \implies T_u \leq \frac{r_u}{q-2}. \end{aligned}$$

□

Example 4.2.2. Take $q = 4, n = 4, X_0 = (1, 2, 3, 2), Y_0 = (4, 2, 3, 2)$, so $S_0 = \{0\}$ and $D_0 = 1$.

$$\begin{aligned} \{X_0(1), X_0(3)\} &= \{2\} = \{Y_0(1), Y_0(3)\} \xrightarrow{3.3.2} L_{X_0}(0) = L_{Y_0}(0) = [4] \setminus \{2\} = \{1, 3, 4\} = L_{X_0}(2) = L_{Y_0}(2) \\ &\xrightarrow{2.2.4} T_0 = T_2 = 1 - 3 \cdot \frac{1}{3} = 0 \\ L_{X_0}(1) &= [4] \setminus \{1, 3\} = \{2, 4\}, \quad L_{Y_0}(1) = [4] \setminus \{4, 3\} = \{1, 2\}, \quad L_{X_0}(1) \cap L_{Y_0}(1) = \{2\} \\ &\xrightarrow{2.2.4} T_1 = 1 - 1 \cdot \min\left\{\frac{1}{2}, \frac{1}{2}\right\} = \frac{1}{2}, \quad L_{X_0}(3) = \{2, 4\}, \quad L_{Y_0}(3) = \{1, 2\} \implies T_3 = \frac{1}{2} \\ &\xrightarrow{4.1.2} \mathbb{E}[D_1 \mid X_0, Y_0] = \frac{1}{4} \sum_{u=0}^3 (D_0 - \mathbf{1}\{u \in S_0\} + T_u) = \frac{1}{4} \left(0 + \frac{3}{2} + 1 + \frac{3}{2}\right) = \frac{4}{4} = 1 \\ \alpha_{\text{cl}}(4, 4) &= 1 - \frac{4-4}{4 \cdot (4-2)} = 1 - 0 = 1 \implies \mathbb{E}[D_1 \mid X_0, Y_0] = \alpha_{\text{cl}}(4, 4) D_0. \end{aligned}$$

4.3. Contraction and mixing on the cycle.

Theorem 4.3.1 (Classical contraction on C_n). *For all $n \geq 3$, $q \geq 3$, and every $(X_0, Y_0) \in \Omega_q(C_n)^2$, the coupling of Definition 4.1.1 satisfies*

$$\mathbb{E}[D_1 \mid X_0, Y_0] \leq \alpha_{\text{cl}}(q, n) D_0, \quad \alpha_{\text{cl}}(q, n) = 1 - \frac{q-4}{n(q-2)},$$

and $\alpha_{\text{cl}}(q, n) < 1$ if and only if $q \geq 5$.

Proof.

$$\begin{aligned} u \sim \text{Unif}(V(C_n)) &\xrightarrow{4.1.2} \mathbb{E}[D_1 \mid X_0, Y_0] = \frac{1}{n} \sum_{u \in V(C_n)} (D_0 - \mathbf{1}\{u \in S_0\} + T_u) = D_0 - \frac{|S_0|}{n} + \frac{1}{n} \sum_{u \in V(C_n)} T_u \\ &\xrightarrow{4.2.1} \sum_{u \in V(C_n)} T_u \leq \frac{1}{q-2} \sum_{u \in V(C_n)} |N(u) \cap S_0| = \frac{1}{q-2} \#\{(u, s) \mid s \in S_0, u \sim s\} = \frac{1}{q-2} \sum_{s \in S_0} \deg(s) = \frac{2D_0}{q-2} \\ |S_0| = D_0 &\implies \mathbb{E}[D_1 \mid X_0, Y_0] \leq D_0 - \frac{D_0}{n} + \frac{2D_0}{n(q-2)} = D_0 \left(1 - \frac{(q-2)-2}{n(q-2)}\right) = D_0 \left(1 - \frac{q-4}{n(q-2)}\right) \\ n(q-2) > 0 &\implies \left(\frac{q-4}{n(q-2)} > 0 \iff q > 4 \iff q \geq 5\right). \end{aligned}$$

□

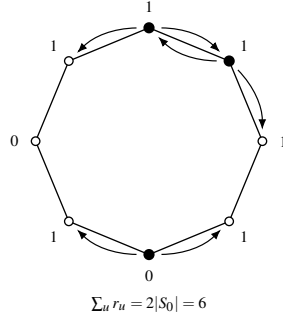


FIGURE 3. S_0 on C_8 (filled); each disagreement charges its two neighbors, and r_u counts arrowheads.

Example 4.3.2. For $n = 8$, $q = 5$, $S_0 = \{0, 1, 4\}$, $D_0 = 3$ as in Figure 3,

$$\begin{aligned} \sum_{u=0}^7 r_u &= 1 + 1 + 1 + 1 + 1 + 0 + 1 + 0 + 1 = 6 = 2|S_0| \xrightarrow{4.3.1} \mathbb{E}[D_1 \mid X_0, Y_0] \leq 3 - \frac{3}{8} + \frac{1}{8} \cdot \frac{6}{5-2} = 3 - \frac{3}{8} + \frac{1}{4} = \frac{23}{8} \\ \alpha_{\text{cl}}(5, 8) &= 1 - \frac{5-4}{8 \cdot (5-2)} = 1 - \frac{1}{24} = \frac{23}{24} \implies \alpha_{\text{cl}}(5, 8) D_0 = \frac{23}{24} \cdot 3 = \frac{23}{8}. \end{aligned}$$

Corollary 4.3.3 (Classical mixing bound on C_n). *For $q \geq 5$,*

$$t_{\text{mix}}(\varepsilon) \leq \left\lceil \frac{n(q-2)}{q-4} (\log(n) + \log(\varepsilon^{-1})) \right\rceil, \quad \mathbb{E}[\tau] \leq \frac{n^2(q-2)}{q-4}.$$

Proof.

$$\begin{aligned} \{D_t = 0\} &= \{X_t = Y_t\}, \quad D_0 \leq n \xrightarrow{2.4.2} t_{\text{mix}}(\varepsilon) \leq \left\lceil \frac{\log(n) + \log(\varepsilon^{-1})}{-\log(\alpha_{\text{cl}}(q, n))} \right\rceil \\ \alpha_{\text{cl}}(q, n) \in (0, 1) &\implies -\log(\alpha_{\text{cl}}(q, n)) \geq 1 - \alpha_{\text{cl}}(q, n) = \frac{q-4}{n(q-2)} \implies \frac{1}{-\log(\alpha_{\text{cl}}(q, n))} \leq \frac{n(q-2)}{q-4} \\ \mathbb{E}[\tau] &= \sum_{t \geq 0} \mathbb{P}(\tau > t) \xrightarrow{2.4.1} \mathbb{E}[\tau] \leq \sum_{t \geq 0} n \alpha_{\text{cl}}(q, n)^t = \frac{n}{1 - \alpha_{\text{cl}}(q, n)} = \frac{n^2(q-2)}{q-4}. \end{aligned}$$

□

Corollary 4.3.4 (The four-cycle). Write $\alpha_{\text{cl}}(q) = \alpha_{\text{cl}}(q, 4)$. For $\varepsilon = 0.05$,

$$\alpha_{\text{cl}}(q) = \frac{3}{4} + \frac{1}{2(q-2)}, \quad \alpha_{\text{cl}}(5) = \frac{11}{12}, \quad \alpha_{\text{cl}}(6) = \frac{7}{8}, \quad \alpha_{\text{cl}}(7) = \frac{17}{20},$$

$$t_{\text{mix}}(0.05) \leq 51, 33, 27, \quad \mathbb{E}[\tau] \leq 48, 32, \frac{80}{3} \quad (q = 5, 6, 7).$$

Proof.

$$1 - \frac{q-4}{4(q-2)} = \frac{4(q-2) - (q-4)}{4(q-2)} = \frac{3q-4}{4(q-2)} = \frac{3(q-2)+2}{4(q-2)} = \frac{3}{4} + \frac{1}{2(q-2)}$$

$$\log\left(\frac{4}{0.05}\right) = \log(80) = 4.382026 \dots \xrightarrow{4.3.3} \left\lceil \frac{\log(80)}{\log(12/11)} \right\rceil = \lceil 50.362 \rceil = 51, \quad \left\lceil \frac{\log(80)}{\log(8/7)} \right\rceil = \lceil 32.816 \rceil = 33,$$

$$\left\lceil \frac{\log(80)}{\log(20/17)} \right\rceil = \lceil 26.963 \rceil = 27, \quad \frac{4^2(q-2)}{q-4} = \frac{16 \cdot 3}{1}, \frac{16 \cdot 4}{2}, \frac{16 \cdot 5}{3} = 48, 32, \frac{80}{3}.$$

□

Remark 4.3.5. Example 4.2.2 attains $\alpha_{\text{cl}}(4, 4) = 1$, and rational enumeration over $\Omega_q(C_4) \times \Omega_q(C_4)$ (Appendix B) gives

$$\max_{(x,y): d_{\text{H}}(x,y)=k} \frac{\mathbb{E}[D_1 \mid X_0 = x, Y_0 = y]}{k} = \alpha_{\text{cl}}(q, 4) \quad \forall k \in \{1, 2, 3, 4\}, \quad q \geq 4,$$

with maximum 1 at $q = 3$.

Remark 4.3.6. The threshold $q \geq 5 = 2\Delta + 1$ is uniform in n and matches the $q > 2\Delta$ regime [10, 11]; going below 2Δ needs different methods [12, 18, 19]. Corollary 4.3.3 gives $t_{\text{mix}}(\varepsilon) = O(n \log(n))$ for fixed $q \geq 5$, and $\alpha_{\text{cl}}(q, n)$ decreases in q .

5. PATH-COUPLING BOUND

Path coupling [6] verifies contraction only on the edge set $\mathcal{E}$ of Hamming-adjacent pairs, at the price, by Proposition 2.5.2, of measuring contraction in the induced path metric $d_{\mathcal{G}}$ of the color graph [21]. This section shows that on $\Omega_q(C_n)$ the substitution $d_{\text{H}} \rightsquigarrow d_{\mathcal{G}}$ is not free. Distances in color graphs are studied in reconfiguration [22, 25, 26]; the obstruction below is its smallest instance.

5.1. Rigid pairs and the metric gap.

Lemma 5.1.1 (Adjacent-difference dichotomy). *Let $x, y \in \Omega_q(C_n)$ differ exactly at adjacent vertices $v \sim u$, let w be the second neighbor of v and z the second neighbor of u . For $q \geq 4$,*

$$d_{\mathcal{G}}(x, y) \in \{2, 3\}, \quad d_{\mathcal{G}}(x, y) = 3 \iff y(v) = x(u) \wedge y(u) = x(v).$$

Proof.

$$\begin{aligned} d_{\mathcal{G}}(x, y) = 2 &\iff y(v) \in L_x(v) \vee y(u) \in L_x(u) \xrightarrow{2.6.2} y(v) \notin \{x(u), x(w)\} \vee y(u) \notin \{x(v), x(z)\} \\ y(v) \neq y(w) = x(w), \quad y(u) \neq y(z) = x(z) &\implies (d_{\mathcal{G}}(x, y) > 2 \iff y(v) = x(u) \wedge y(u) = x(v)) \\ y(v) = x(u), y(u) = x(v), q \geq 4 &\implies \exists s \in [q] \setminus \{x(v), x(u), x(w)\} \\ \implies x \xrightarrow{v \mapsto s} x' \xrightarrow{u \mapsto x(v)} x'' \xrightarrow{v \mapsto x(u)} y, \quad s \notin \{x(u), x(w)\}, \quad x(v) \notin \{s, x(z)\}, \quad x(u) \notin \{x(v), x(w)\} \\ &\implies d_{\mathcal{G}}(x, y) \leq 3 \xrightarrow{2.7.2} d_{\mathcal{G}}(x, y) \in \{2, 3\}. \end{aligned}$$

□

Proposition 5.1.2 (Swap pairs are rigid). *Let $x, y \in \Omega_q(C_n)$ differ exactly at adjacent vertices $v \sim u$ with*

$$y(v) = x(u), \quad y(u) = x(v).$$

Then

$$d_{\text{H}}(x, y) = 2, \quad d_{\mathcal{G}}(x, y) \geq 3, \quad d_{\mathcal{G}}(x, y) = 3 \quad (q \geq 4).$$

Such pairs exist in $\Omega_q(C_n)$ for every $n \geq 3$ and $q \geq 4$; on C_4 they exist if and only if $q \geq 4$.

Proof.

$$\begin{aligned}
u \in N(v) &\xrightarrow{2.6.2} y(v) = x(u) \notin L_x(v), \quad v \in N(u) \implies y(u) = x(v) \notin L_x(u) \xrightarrow{2.7.2} d_{\mathcal{G}}(x, y) > 2 \\
q \geq 4 &\xrightarrow{5.1.1} d_{\mathcal{G}}(x, y) = 3 \\
w \in N(v) \setminus \{u\}, \quad z \in N(u) \setminus \{v\}, \quad x, y \text{ proper} &\iff x(w) \notin \{x(v), x(u)\}, \quad x(z) \notin \{x(v), x(u)\} \\
n = 4 \implies w \sim z \implies x(w) \neq x(z) \implies |[q] \setminus \{x(v), x(u)\}| \geq 2 &\iff q \geq 4, \quad n \neq 4 \implies w \not\sim z.
\end{aligned}$$

□

Example 5.1.3. For $q = 5, n = 4$, the pair $x = (0, 1, 2, 3), y = (1, 0, 2, 3)$ is a swap pair at the edge $(0, 1)$, drawn in Figure 4.

$$\begin{aligned}
y(0) = 1 = x(1) &\xrightarrow{2.6.2} 1 \notin L_x(0) = \{0, 2, 4\}, \quad y(1) = 0 = x(0) \implies 0 \notin L_x(1) = \{1, 3, 4\} \\
&\xrightarrow{5.1.2} d_{\mathcal{G}}(x, y) = 3, \quad (0, 1, 2, 3) \xrightarrow{v_0 \mapsto 4} (4, 1, 2, 3) \xrightarrow{v_1 \mapsto 0} (4, 0, 2, 3) \xrightarrow{v_0 \mapsto 1} (1, 0, 2, 3),
\end{aligned}$$

each intermediate proper since $4 \notin \{1, 3\}, 0 \notin \{4, 2\}, 1 \notin \{0, 3\}$. Breadth-first search (Appendix B) finds 480 swap pairs in $\Omega_5(C_4)$ and $\text{diam } d_{\mathcal{G}} = 6 > 4 = \text{diam } d_H$.

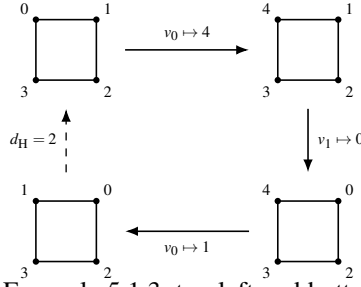


FIGURE 4. The swap pair of Example 5.1.3, top left and bottom left. The dashed edge is the direct route, which no single recoloring realizes; the three solid moves close the square, so $d_{\mathcal{G}} = 3$.

5.2. Edge contraction in the path metric.

Theorem 5.2.1 (Path-coupling contraction on C_n). *For all $n \geq 3, q \geq 4$, and every color-graph edge (x, y) , the coupling of Definition 4.1.1 satisfies*

$$\mathbb{E}[d_{\mathcal{G}}(X_1, Y_1) \mid X_0 = x, Y_0 = y] \leq \alpha_{\text{pc}}(q, n) d_{\mathcal{G}}(x, y), \quad \alpha_{\text{pc}}(q, n) = 1 - \frac{q-6}{n(q-2)},$$

and $\alpha_{\text{pc}}(q, n) < 1$ if and only if $q \geq 7$.

Proof.

$$\begin{aligned}
S_0 = \{v\}, \quad u = v &\implies r_v = |N(v) \cap \{v\}| = 0 \xrightarrow{4.2.1} T_v = 0 \xrightarrow{4.1.2} D_1 = 0 \implies d_{\mathcal{G}}(X_1, Y_1) = 0 \\
u \notin \{v\} \cup N(v) &\implies r_u = |N(u) \cap \{v\}| = 0 \implies T_u = 0 \implies d_{\mathcal{G}}(X_1, Y_1) = 1 \quad (\#\{u\} = n-3) \\
u \in N(v) &\xrightarrow{4.1.2} \mathbb{P}(d_{\mathcal{G}}(X_1, Y_1) = 1) = 1 - T_u, \quad X_1 \neq Y_1 \text{ at } \{v, u\}, \quad v \sim u \xrightarrow{5.1.1} d_{\mathcal{G}}(X_1, Y_1) \leq 3 \\
&\xrightarrow{4.2.1} \mathbb{E}[d_{\mathcal{G}}(X_1, Y_1) \mid u \in N(v)] \leq (1 - T_u) \cdot 1 + T_u \cdot 3 = 1 + 2T_u \leq 1 + \frac{2}{q-2} \\
&\implies \mathbb{E}[d_{\mathcal{G}}(X_1, Y_1)] \leq \frac{1}{n} \left(0 + (n-3) \cdot 1 + 2 \cdot \left(1 + \frac{2}{q-2} \right) \right) = 1 - \frac{1}{n} + \frac{4}{n(q-2)} = 1 - \frac{q-6}{n(q-2)} \\
n(q-2) > 0 &\implies \left(\frac{q-6}{n(q-2)} > 0 \iff q > 6 \iff q \geq 7 \right).
\end{aligned}$$

□

Example 5.2.2 (The bound is attained). For $q = 5$, $n = 4$, take the color-graph edge $x = (0, 1, 2, 3)$, $y = (4, 1, 2, 3)$, differing at $v = 0$.

$$\begin{aligned}
u \in \{0, 2\} &\xrightarrow{3,3,2} T_0 = T_2 = 0 \implies d_{\mathcal{G}}(X_1, Y_1) = 0 \ (u = 0), \quad d_{\mathcal{G}}(X_1, Y_1) = 1 \ (u = 2) \\
u = 1 &\implies L_x(1) = [5] \setminus \{0, 2\} = \{1, 3, 4\}, \quad L_y(1) = [5] \setminus \{4, 2\} = \{0, 1, 3\} \xrightarrow{2,2,4} T_1 = 1 - 2 \cdot \frac{1}{3} = \frac{1}{3} \\
&U_{L_x(1)} - \min\{U_{L_x(1)}, U_{L_y(1)}\} = \frac{1}{3} \delta_4, \quad U_{L_y(1)} - \min\{U_{L_x(1)}, U_{L_y(1)}\} = \frac{1}{3} \delta_0 \\
&\implies (X_1(1), Y_1(1)) = (4, 0) \text{ on the mismatch,} \quad ((0, 4, 2, 3), (4, 0, 2, 3)) \text{ a swap of } \{0, 4\} \xrightarrow{5,1,2} d_{\mathcal{G}} = 3 \\
u = 3 &\implies T_3 = \frac{1}{3}, \quad ((0, 1, 2, 4), (4, 1, 2, 0)) \text{ a swap across } (3, 0) \implies d_{\mathcal{G}} = 3 \\
&\implies \mathbb{E}[d_{\mathcal{G}}(X_1, Y_1)] = \frac{1}{4} \left(0 + 1 + \left(\frac{2}{3} \cdot 1 + \frac{1}{3} \cdot 3 \right) + \left(\frac{2}{3} \cdot 1 + \frac{1}{3} \cdot 3 \right) \right) = \frac{1}{4} \left(0 + 1 + \frac{5}{3} + \frac{5}{3} \right) = \frac{13}{12} = \alpha_{\text{pc}}(5, 4).
\end{aligned}$$

Corollary 5.2.3 (The four-cycle). Write $\alpha_{\text{pc}}(q) = \alpha_{\text{pc}}(q, 4)$. For $\varepsilon = 0.05$,

$$\begin{aligned}
\alpha_{\text{pc}}(q) &= \frac{3}{4} + \frac{1}{q-2}, \quad \alpha_{\text{pc}}(7) = \frac{19}{20}, \quad \alpha_{\text{pc}}(8) = \frac{11}{12}, \\
t_{\text{mix}}(0.05) &\leq \left\lceil \frac{\log(120)}{\log(20/19)} \right\rceil = \lceil 93.335 \rceil = 94, \quad \left\lceil \frac{\log(120)}{\log(12/11)} \right\rceil = \lceil 55.024 \rceil = 56 \quad (q = 7, 8).
\end{aligned}$$

Proof.

$$\begin{aligned}
1 - \frac{q-6}{4(q-2)} &= \frac{4(q-2) - (q-6)}{4(q-2)} = \frac{3q-2}{4(q-2)} = \frac{3(q-2)+4}{4(q-2)} = \frac{3}{4} + \frac{1}{q-2} \\
\text{diam } d_{\mathcal{G}} = 6 \text{ (Appendix B)} &\xrightarrow{2.5,3} t_{\text{mix}}(0.05) \leq \left\lceil \frac{\log(6) + \log(20)}{-\log(\alpha_{\text{pc}}(q))} \right\rceil, \quad \log(120) = 4.787492\dots
\end{aligned}$$

□

Corollary 5.2.4 (Separation). For all $n \geq 3$ and $q \geq 4$,

$$\alpha_{\text{pc}}(q, n) - \alpha_{\text{cl}}(q, n) = \frac{(q-4) - (q-6)}{n(q-2)} = \frac{2}{n(q-2)} > 0,$$

so the drift analysis contracts for $q \geq 5$ while path coupling on Hamming-adjacent edges contracts only for $q \geq 7$, uniformly in n , as Corollary 2.5.4 predicts with $\rho = d_{\text{H}}$ and $d_{\mathcal{E}} = d_{\mathcal{G}}$.

Remark 5.2.5 (Sharpness). Example 5.2.2 attains $\alpha_{\text{pc}}(5, 4)$, and rational enumeration over all color-graph edges (Appendix B) gives

$$\max_{(x,y): d_{\mathcal{G}}(x,y)=1} \mathbb{E}[d_{\mathcal{G}}(X_1, Y_1)] = \alpha_{\text{pc}}(q, 4) \quad \forall q \geq 4,$$

with maximum 1 at $q = 3$. The two constants differ only in the charge for a created disagreement, +1 for d_{H} against +2 for $d_{\mathcal{G}}$, and the +2 is exact because the mismatch in Example 5.2.2 lands deterministically on a swap pair. Whether a weighted edge metric restores $q \geq 5$ under edge-only verification is open; metric choice for coloring dynamics is active [8, 9, 20].

6. COMPARISON, EXACT COMPUTATION, AND CLOSED FORMS

6.1. Separation of the two bounds.

Corollary 6.1.1 (Thresholds). For all $n \geq 3$ and $q \geq 4$,

$$\alpha_{\text{cl}}(q, n) < 1 \iff q \geq 5, \quad \alpha_{\text{pc}}(q, n) < 1 \iff q \geq 7,$$

and for $q \geq 7$,

$$\frac{n(q-2)}{q-4} (\log(n) + \log(\varepsilon^{-1})) < \frac{n(q-2)}{q-6} (\log(\text{diam } d_{\mathcal{G}}) + \log(\varepsilon^{-1})).$$

Proof.

$$1 - \alpha_{\text{cl}}(q, n) = \frac{q-4}{n(q-2)} > 0 \iff q > 4, \quad 1 - \alpha_{\text{pc}}(q, n) = \frac{q-6}{n(q-2)} > 0 \iff q > 6$$

$$0 < q-6 < q-4 \xrightarrow{5.2.4} \frac{n(q-2)}{q-4} < \frac{n(q-2)}{q-6} \xrightarrow{2.7.2} n = \text{diam} d_{\text{H}} \leq \text{diam} d_{\mathcal{G}}.$$

□

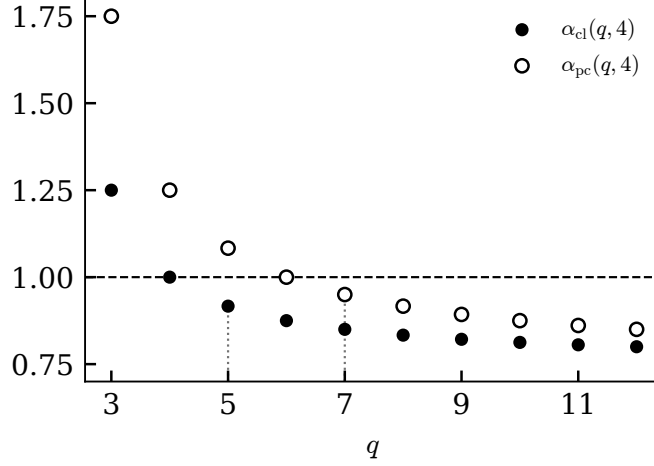


FIGURE 5. $\alpha_{\text{cl}}(q, 4)$ (filled) and $\alpha_{\text{pc}}(q, 4)$ (open); dotted verticals at the thresholds $q = 5, 7$, dashed line at 1.

Example 6.1.2. For $q = 7, n = 4, \varepsilon = 0.05$,

$$\alpha_{\text{cl}}(7) = \frac{3}{4} + \frac{1}{10} = \frac{17}{20}, \quad \alpha_{\text{pc}}(7) = \frac{3}{4} + \frac{1}{5} = \frac{19}{20} \xrightarrow{5.2.4} \alpha_{\text{pc}}(7) - \alpha_{\text{cl}}(7) = \frac{2}{20} = \frac{2}{4 \cdot 5}$$

$$\left\lceil \frac{\log(80)}{\log(20/17)} \right\rceil = \lceil 26.963 \rceil = 27, \quad \left\lceil \frac{\log(120)}{\log(20/19)} \right\rceil = \lceil 93.335 \rceil = 94, \quad 18 < 27 < 94.$$

Example 6.1.3.

$$\lim_{q \rightarrow \infty} \alpha_{\text{cl}}(q, n) = \lim_{q \rightarrow \infty} \alpha_{\text{pc}}(q, n) = 1 - \frac{1}{n}, \quad \alpha_{\text{pc}}(q, n) - \alpha_{\text{cl}}(q, n) = \frac{2}{n(q-2)} \rightarrow 0,$$

$$n = 4, q = 12 \implies \alpha_{\text{cl}} = \frac{4}{5}, \quad \alpha_{\text{pc}} = \frac{17}{20}, \quad \alpha_{\text{pc}} - \alpha_{\text{cl}} = \frac{1}{20} = \frac{2}{4 \cdot 10}.$$

6.2. The diameter of the color graph.

Definition 6.2.1. A *schedule* from x to y is a sequence $x = z_0, z_1, \dots, z_L = y$ in $\Omega_q(C_n)$ with $(z_{j-1}, z_j) \in E(\mathcal{C}_q(C_n))$ for every j , and

$$m_v = \#\{j \mid z_j(v) \neq z_{j-1}(v)\}, \quad L = \sum_{v \in V(C_n)} m_v.$$

Lemma 6.2.2 (Charge). Let $e_1, \dots, e_s \in E(C_n)$ satisfy $e_i \cap e_j = \emptyset$ for $i \neq j$, and for each $e_i = (v, u)$

$$y(v) = x(u) \neq x(v), \quad y(u) = x(v) \neq x(u).$$

Then every schedule from x to y satisfies $L \geq d_{\text{H}}(x, y) + s$.

Proof.

$$x(v) \neq y(v) \implies m_v \geq 1 \implies \sum_v m_v \geq d_{\text{H}}(x, y)$$

$$e_i = (v, u), \quad t_v = \max\{j \mid z_j(v) \neq z_{j-1}(v)\} < t_u, \quad z_{t_v}(v) = y(v) \xrightarrow{2.6.1} z_{t_v}(u) \neq y(v) = x(u)$$

$$\implies \exists j < t_v \quad z_j(u) \neq z_{j-1}(u) \implies m_u \geq 2$$

$$e_i \cap e_j = \emptyset \implies \sum_v m_v \geq d_H(x, y) + s.$$

□

Theorem 6.2.3 (Diameter on C_n). *For all $n \geq 3$ and $q \geq 5$, with $m = \lfloor n/2 \rfloor$,*

$$\text{diam } d_{\mathcal{G}} = \left\lfloor \frac{3n}{2} \right\rfloor = \text{diam } d_H + m.$$

Proof.

$$\begin{aligned} x|_{\{0, \dots, 2m-1\}} &= (0, 1, \dots, 0, 1), & y|_{\{0, \dots, 2m-1\}} &= (1, 0, \dots, 1, 0), & 2 \nmid n &\implies (x(n-1), y(n-1)) = (2, 3) \\ \implies d_H(x, y) &= n, & y(2i) &= x(2i+1), & y(2i+1) &= x(2i), & (2i, 2i+1) \cap (2j, 2j+1) &= \emptyset \ (i \neq j) \end{aligned}$$

$$\xRightarrow{6.2.2} \text{diam } d_{\mathcal{G}} \geq n + m = \left\lfloor \frac{3n}{2} \right\rfloor.$$

$$I \subset V(C_n) \text{ independent, } |I| = m, \quad 2 \nmid n \implies \{(a, a+1)\} = E(C_n[V \setminus I])$$

$$\begin{aligned} x(v+1) = y(v) \wedge x(v) = y(v+1) \ \forall v &\implies x(v+2) = y(v+1) = x(v) \ \forall v \xrightarrow{\gcd(2,n)=1} x(0) = x(1) \xrightarrow{2.6.1} \perp \\ &\implies \exists \text{ rotation of } I \ x(a+1) \neq y(a) \end{aligned}$$

$$\begin{aligned} \text{(i)} \ z(v) &:= s_v \in [q] \setminus \{x(v-1), x(v+1), y(v-1), y(v+1)\} \ (v \in I), & q \geq 5 &\implies s_v \text{ exists} \\ \text{(ii)} \ z(a) &:= y(a), \quad z(a+1) := y(a+1), \quad z(u) := y(u) \ (u \notin I \cup \{a, a+1\}), & \text{(iii)} \ z(v) &:= y(v) \ (v \in I) \\ \text{(i)} \ s_v &\notin \{x(v-1), x(v+1)\}, & \text{(ii)} \ u \pm 1 \in I &\implies y(u) \notin \{s_{u-1}, s_{u+1}\}, \\ y(a) &\notin \{s_{a-1}, x(a+1)\}, & y(a+1) &\notin \{y(a), s_{a+2}\}, & \text{(iii)} \ y(v) &\notin \{y(v-1), y(v+1)\} \\ \implies d_{\mathcal{G}}(x, y) &\leq |I| + |V \setminus I| + |I| = m + (n - m) + m = \left\lfloor \frac{3n}{2} \right\rfloor. \end{aligned}$$

□

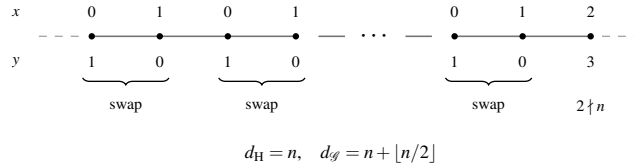


FIGURE 6. The extremal pair of Theorem 6.2.3 on C_n .

Example 6.2.4. For $n = 5, q = 5$, the extremal pair $x = (0, 1, 0, 1, 2), y = (1, 0, 1, 0, 3)$ has $I = \{1, 3\}$, leftover edge $(4, 0)$ with $x(0) = 0 \neq 3 = y(4)$, and

$$s_1 \in [5] \setminus \{0, 0, 1, 1\}, \quad s_3 \in [5] \setminus \{0, 2, 1, 3\} \implies s_1 = s_3 = 4.$$

Figure 7 records the schedule row by row, and

$$L = 2 + 3 + 2 = 7 \xRightarrow{6.2.2} 7 \leq d_{\mathcal{G}}(x, y) \leq L \implies d_{\mathcal{G}}(x, y) = 7 = \left\lfloor \frac{15}{2} \right\rfloor.$$

t	v_0	v_1	v_2	v_3	v_4	move
0	0	1	0	1	2	
1	0	4	0	1	2	$v_1 \mapsto s_1$
2	0	4	0	4	2	$v_3 \mapsto s_3$
3	1	4	0	4	2	$v_0 \mapsto y(v_0)$
4	1	4	1	4	2	$v_2 \mapsto y(v_2)$
5	1	4	1	4	3	$v_4 \mapsto y(v_4)$
6	1	0	1	4	3	$v_1 \mapsto y(v_1)$
7	1	0	1	0	3	$v_3 \mapsto y(v_3)$

FIGURE 7. Space-time record of the three-phase schedule of Example 6.2.4; boxes mark moves.

Example 6.2.5. For $n = 6$, $q \geq 5$, $x = (0, 1, 0, 1, 0, 1)$, $y = (1, 0, 1, 0, 1, 0)$, $I = \{0, 2, 4\}$, $s_0 = s_2 = s_4 = 2$,

$$(0, 1, 0, 1, 0, 1) \xrightarrow{(i)} (2, 1, 2, 1, 2, 1) \xrightarrow{(ii)} (2, 0, 2, 0, 2, 0) \xrightarrow{(iii)} (1, 0, 1, 0, 1, 0),$$

$$L = 3 + 3 + 3 = 9 \xrightarrow{6.2.2} 9 \leq d_{\mathcal{G}}(x, y) \leq L \implies d_{\mathcal{G}}(x, y) = 9 = \left\lfloor \frac{18}{2} \right\rfloor.$$

Example 6.2.6. Breadth-first search (Appendix B) confirms $\text{diam } d_{\mathcal{G}} = \lfloor 3n/2 \rfloor$ at every computed pair, including $q = 4$, where Theorem 6.2.3 asserts only the lower bound,

(n, q)	(3, 4)	(3, 5)	(4, 3)	(4, 4)	(4, 5)	(5, 4)	(5, 5)	(6, 4)	(7, 4)
$\text{diam } d_{\mathcal{G}}$	4	4	6	6	6	7	7	9	10
$\lfloor 3n/2 \rfloor$	4	4	6	6	6	7	7	9	10

Remark 6.2.7. Phase (i) is the only use of the fifth color, so for $q = 4$ the theorem gives the lower bound and Example 6.2.6 the value for $n \leq 7$. Color-graph distances are superpolynomial below $\Delta + 2$ colors [22], with diameters surveyed in [25, 26]; the closed form here rests on the one-dimensional geometry of the cycle.

6.3. Exact computation on the four-cycle. Since $|\Omega_q(C_4)| \leq 1302$ for $q \leq 7$, every quantity is computed exactly. The transition matrix of Definition 2.6.1 is assembled over $\Omega_q(C_4)$, contraction constants are evaluated in rational arithmetic over all pairs, and $d_{\mathcal{G}}$ by breadth-first search in $\mathcal{C}_q(C_4)$; Appendix B.

q	$ \Omega_q(C_4) $	$\alpha_{\text{cl}}(q)$	$\alpha_{\text{pc}}(q)$	classical	path	$t_{\text{mix}}(0.05)$	λ_*
3	18	5/4	7/4	—	—	57	0.9557
4	84	1	5/4	—	—	29	0.9115
5	260	11/12	13/12	51	—	22	0.8734
6	630	7/8	1	33	—	19	0.8494
7	1302	17/20	19/20	27	94	18	0.8330

TABLE 1. Exact data for C_4 at $\varepsilon = 0.05$; dashes mark absent contraction.

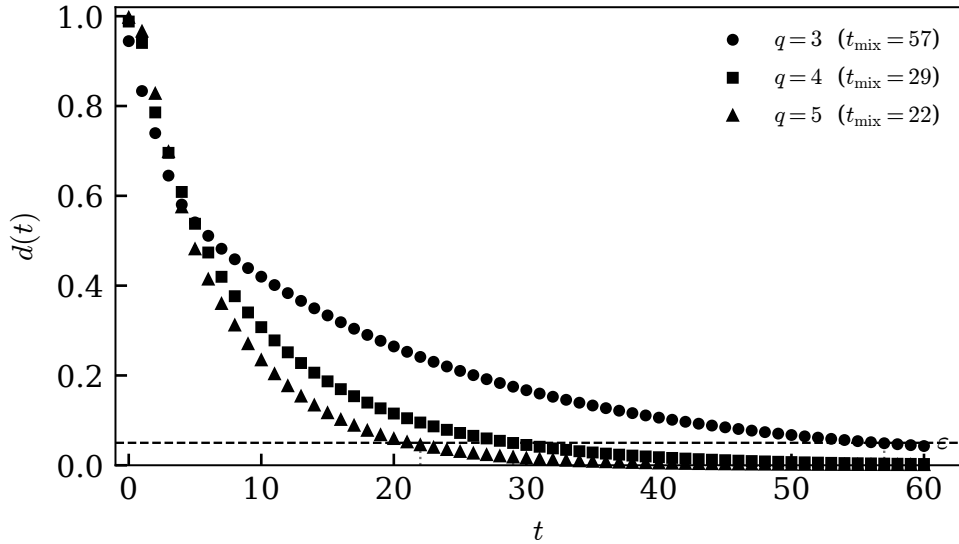


FIGURE 8. Exact $d(t) = \max_{x \in \Omega_q(C_4)} \|P^t(x, \cdot) - \pi\|_{\text{TV}}$; dotted verticals at $t_{\text{mix}}(0.05) = 22, 29, 57$, dashed line at $\varepsilon = 0.05$.

Proposition 6.3.1 (Both constants are attained). *For every $q \geq 4$,*

$$\max_{\substack{(x,y) \in \Omega_q(C_4)^2 \\ d_H(x,y)=k}} \frac{\mathbb{E}[D_1 \mid X_0 = x, Y_0 = y]}{k} = \alpha_{cl}(q, 4) \quad \forall k \in \{1, 2, 3, 4\}, \quad \max_{\substack{(x,y) \in \Omega_q(C_4)^2 \\ d_{\mathcal{G}}(x,y)=1}} \mathbb{E}[d_{\mathcal{G}}(X_1, Y_1)] = \alpha_{pc}(q, 4),$$

and both maxima equal 1 at $q = 3$.

Proof.

rational enumeration over $\Omega_q(C_4) \times \Omega_q(C_4)$ (Appendix B) $\xrightarrow{4.2.2, 5.2.2}$ attainment at $q = 4, 5$. □

Proposition 6.3.2 (Geometry of the color graph on C_4). *For every $q \geq 3$,*

$$\begin{aligned} \text{diam } d_H &= 4, & \text{diam } d_{\mathcal{G}} &= 6, & \max_{x,y} (d_{\mathcal{G}}(x,y) - d_H(x,y)) &= 2, \\ \#\{(x,y) \mid d_H &= 2, d_{\mathcal{G}} = 3\} &= 0, 96, 480, 1440, 3360 & & (q = 3, \dots, 7). \end{aligned}$$

Proof.

$$q \geq 5 \xrightarrow{6.2.3} \text{diam } d_{\mathcal{G}} = \left\lfloor \frac{3 \cdot 4}{2} \right\rfloor = 6, \quad q \in \{3, 4\} \xrightarrow{6.2.6} \text{diam } d_{\mathcal{G}} = 6$$

breadth-first search (Appendix B) $\xrightarrow{5.1.2}$ the counted pairs are the swap pairs, absent at $q = 3$. □

Remark 6.3.3 (Resolution of sampled estimates). For $\hat{\mu}_N$ the empirical measure of N trajectories and π uniform on $|\Omega_q(C_4)|$ atoms,

$$\begin{aligned} \text{Var}(\hat{\mu}_N(z)) &= \frac{\pi(z)(1-\pi(z))}{N} \implies \mathbb{E}|\hat{\mu}_N(z) - \pi(z)| \leq \sqrt{\frac{\pi(z)}{N}} \\ \implies \mathbb{E}\|\hat{\mu}_N - \pi\|_1 &\leq \sum_{z \in \Omega_q(C_4)} \sqrt{\frac{1}{|\Omega_q(C_4)|N}} = \sqrt{\frac{|\Omega_q(C_4)|}{N}} = 0.134, 0.290, 0.510, 0.794, 1.141 \quad (N = 10^3, q = 3, \dots, 7), \end{aligned}$$

exceeding the range of $d(t)$ once $q \geq 6$. A sampled curve therefore plateaus at a level increasing in q , an artifact of $|\Omega_q(C_4)|$, whereas Table 1 gives $t_{\text{mix}}(0.05)$ strictly decreasing in q .

6.4. Spectral analysis.

Proposition 6.4.1 (Spectral bound for the symmetric chain). *Let $N = |\Omega_q(C_4)|$ and $P = P^\top$ as in Lemma 2.6.3. Then there exist $Q \in \mathbb{R}^{N \times N}$ and real $1 = \lambda_1 \geq \dots, \lambda_\star = \max_{k \geq 2} (|\lambda_k|)$, with*

$$Q^\top Q = QQ^\top = I_N = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}, \quad P = Q\Lambda Q^\top, \quad \Lambda = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_N \end{bmatrix}, \quad Q_{x1} = \frac{1}{\sqrt{N}},$$

and

$$d(t) \leq \frac{1}{2} \sqrt{N} \lambda_\star^t \quad \text{for all } t \geq 0.$$

Proof.

$$\begin{aligned} P = P^\top &\implies P = Q\Lambda Q^\top, \quad Q^\top Q = I_N \quad (\text{spectral theorem for real symmetric matrices}) \\ \sum_y P(x,y) &= 1 \quad \forall x, \quad P = P^\top \implies P\mathbf{1} = \mathbf{1}, \quad \mathbf{1} = (1, 1, \dots, 1)^\top \implies \lambda_1 = 1, \quad Qe_1 = \frac{\mathbf{1}}{\sqrt{N}} \\ P^2 &= (Q\Lambda Q^\top)(Q\Lambda Q^\top) = Q\Lambda(Q^\top Q)\Lambda Q^\top = Q\Lambda I_N \Lambda Q^\top = Q\Lambda^2 Q^\top \implies P^t = Q\Lambda^t Q^\top \quad \forall t \\ P^t(x,y) &= \sum_{k=1}^N \lambda_k^t Q_{xk} Q_{yk} = \frac{1}{N} + \sum_{k=2}^N \lambda_k^t Q_{xk} Q_{yk} \implies P^t(x,y) - \pi(y) = \sum_{k=2}^N \lambda_k^t Q_{xk} Q_{yk} \\ 2\|P^t(x, \cdot) - \pi\|_{\text{TV}} &= \sum_y \left| \sum_{k=2}^N \lambda_k^t Q_{xk} Q_{yk} \right| \leq \sqrt{N} \left(\sum_y \left(\sum_{k=2}^N \lambda_k^t Q_{xk} Q_{yk} \right)^2 \right)^{1/2} \end{aligned}$$

$$\begin{aligned} \sum_y \left(\sum_{k=2}^N \lambda_k^t Q_{xk} Q_{yk} \right)^2 &= \sum_{k=2}^N \sum_{l=2}^N \lambda_k^t \lambda_l^t Q_{xk} Q_{xl} \sum_y Q_{yk} Q_{yl} = \sum_{k=2}^N \sum_{l=2}^N \lambda_k^t \lambda_l^t Q_{xk} Q_{xl} (I_N)_{kl} = \sum_{k=2}^N \lambda_k^{2t} Q_{xk}^2 \\ &\leq \lambda_\star^{2t} \sum_{k=1}^N Q_{xk}^2 = \lambda_\star^{2t} (QQ^\top)_{xx} = \lambda_\star^{2t} (I_N)_{xx} = \lambda_\star^{2t} \implies d(t) \leq \frac{1}{2} \sqrt{N} \lambda_\star^t. \end{aligned}$$

□

Example 6.4.2 (Drift and spectral bounds against the exact decay). For $q = 5$, $N = 260$, $\alpha_{cl}(5) = \frac{11}{12}$, $\lambda_\star = 0.873446$,

$$d(10) = 0.2335 < 4\left(\frac{11}{12}\right)^{10} = 1.6756, \quad d(22) = 0.0444 < 4\left(\frac{11}{12}\right)^{22} = 0.5899, \quad d(30) = 0.0150 < 0.2941,$$

$$\frac{1}{2} \sqrt{260} = 8.0623, \quad (0.873446)^{22} = \exp(22 \log(0.873446)) = \exp(-2.977) = 0.05090$$

$$\xrightarrow{6.4.1} d(22) \leq 8.0623 \times 0.05090 = 0.4104,$$

and $\lambda_\star < \alpha_{cl}(q)$ for every q in Table 1.

6.5. Closed forms.

Theorem 6.5.1 (Both methods on C_n , closed form). *For every $n \geq 3$ and the coupling of Definition 4.1.1,*

$$q \geq 5 \implies t_{\text{mix}}(\varepsilon) \leq \left\lceil \frac{n(q-2)}{q-4} (\log(n) + \log(\varepsilon^{-1})) \right\rceil,$$

$$q \geq 7 \implies t_{\text{mix}}(\varepsilon) \leq \left\lceil \frac{n(q-2)}{q-6} \left(\log\left(\left\lfloor \frac{3n}{2} \right\rfloor\right) + \log(\varepsilon^{-1}) \right) \right\rceil,$$

with the first strictly smaller for every $q \geq 7$, and both $O(n \log(n))$ for fixed q .

Proof.

$$\text{diam } d_H = n \xrightarrow{4.3.3} \text{ the first bound,} \quad q \geq 7 \xrightarrow{6.2.3} \text{diam } d_{\mathcal{G}} = \left\lfloor \frac{3n}{2} \right\rfloor \xrightarrow{2.5.3} \text{ the second}$$

$$0 < q-6 < q-4, \quad n \leq \left\lfloor \frac{3n}{2} \right\rfloor \xrightarrow{6.1.1} \text{ the ordering,} \quad \log\left(\left\lfloor \frac{3n}{2} \right\rfloor\right) \leq \log(n) + \log(2).$$

□

Example 6.5.2. For $n = 6$, $q = 7$, $\varepsilon = 0.05$,

$$\frac{6 \cdot 5}{3} (\log(6) + \log(20)) = 10 \times 4.787492 = 47.875 \implies t_{\text{mix}}(0.05) \leq 48,$$

$$\left\lfloor \frac{3 \cdot 6}{2} \right\rfloor = 9, \quad \frac{6 \cdot 5}{1} (\log(9) + \log(20)) = 30 \times 5.192957 = 155.789 \implies t_{\text{mix}}(0.05) \leq 156, \quad \frac{156}{48} = 3.25.$$

Theorem 6.5.3 (The four-cycle, closed form). *On $\Omega_q(C_4)$,*

$$|\Omega_q(C_4)| = (q-1)^4 + (q-1), \quad \alpha_{cl}(q) = \frac{3}{4} + \frac{1}{2(q-2)}, \quad \alpha_{pc}(q) = \frac{3}{4} + \frac{1}{q-2},$$

both attained, with thresholds $q \geq 5$ and $q \geq 7$, and

$$\text{diam } d_H = 4, \quad \text{diam } d_{\mathcal{G}} = 6, \quad t_{\text{mix}}(0.05) = 57, 29, 22, 19, 18 \quad (q = 3, \dots, 7).$$

Proof.

$$\begin{array}{lll} \text{Proposition 3.1.1,} & \text{Theorems 4.3.1, 5.2.1 at } n = 4, & \text{Proposition 6.3.1,} \\ & \text{Proposition 6.3.2,} & \text{Table 1.} \end{array}$$

□

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APPENDIX A. FOUNDATIONAL TOOLS

A.1. Linear algebra over $\mathbb{R}$.

Definition A.1.1. $A \in \text{Mat}_{N \times N}(\mathbb{R})$ is *symmetric* if $A = A^\top$, and $Q \in \text{Mat}_{N \times N}(\mathbb{R})$ is *orthogonal* if $Q^\top Q = QQ^\top = I_N$.

Theorem A.1.2 (Spectral theorem, real symmetric case). *Let $A = A^\top \in \text{Mat}_{N \times N}(\mathbb{R})$. Then every eigenvalue of A is real, and there exist an orthogonal Q and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_N)$, $\lambda_j \in \mathbb{R}$, with*

$$A = Q\Lambda Q^\top.$$

Proof.

$$\begin{aligned} A\xi = \lambda\xi, \quad 0 \neq \xi \in \mathbb{C}^N &\implies \lambda(\xi^* \xi) = \xi^*(A\xi) = (A\xi)^* \xi = (\lambda\xi)^* \xi = \bar{\lambda}(\xi^* \xi) \implies (\lambda - \bar{\lambda})(\xi^* \xi) = 0 \implies \lambda \in \mathbb{R} \\ N = 1 &\implies A = (\lambda_1), \quad N \geq 2, \quad A\xi_1 = \lambda_1 \xi_1, \quad \|\xi_1\| = 1, \quad W = (\xi_1)^\perp \subset \mathbb{R}^N \\ w \in W &\implies \xi_1^\top (Aw) = (A\xi_1)^\top w = \lambda_1(\xi_1^\top w) = 0 \implies AW \subseteq W, \quad (A|_W)^\top = A|_W \\ &\stackrel{\text{induction}}{\implies} \exists \text{ orthonormal eigenbasis } \xi_1, \dots, \xi_N \implies Q = (\xi_1 \cdots \xi_N), \quad AQ = Q\Lambda \implies A = Q\Lambda Q^\top. \end{aligned}$$

□

Corollary A.1.3. *If additionally $A_{xy} \geq 0$ and $\sum_y A_{xy} = 1$ for every x , then every eigenvalue satisfies $\lambda \in [-1, 1]$.*

Proof.

$$A\xi = \lambda\xi, \quad |\xi_i| = \max_j (|\xi_j|) > 0 \implies |\lambda| |\xi_i| = \left| \sum_j A_{ij} \xi_j \right| \leq \sum_j A_{ij} |\xi_j| \leq |\xi_i| \sum_j A_{ij} = |\xi_i| \implies |\lambda| \leq 1.$$

□

APPENDIX B. REPRODUCIBILITY

The chain is assembled entrywise from Definition 2.6.1,

$$\begin{aligned} P(x, y) &= \frac{1}{4} \cdot \frac{1}{|L_x(v)|} \text{ if } x, y \text{ differ exactly at } v \text{ and } y(v) \in L_x(v), \quad P(x, x) = 1 - \sum_{y \neq x} P(x, y), \\ d(t) &= \max_{x \in \Omega_q(C_4)} \frac{1}{2} \sum_{y \in \Omega_q(C_4)} \left| (P^t)(x, y) - \frac{1}{|\Omega_q(C_4)|} \right|, \quad \lambda_* = \max_{k \geq 2} (|\lambda_k|) \stackrel{\text{A.1.2}}{\implies} \lambda_k \in \mathbb{R}. \end{aligned}$$

One-step expectations are exact rationals via Lemmas 4.1.2 and 2.2.4, the update at u being coupled by

$$\begin{aligned} \mu &= U_{L_x(u)}, \quad \nu = U_{L_y(u)}, \quad r_\mu = \mu - \min\{\mu, \nu\}, \quad r_\nu = \nu - \min\{\mu, \nu\}, \\ \mathbb{P}(X_1(u) = a, Y_1(u) = b) &= \min\{\mu(a), \nu(a)\} \mathbf{1}\{a = b\} + \frac{r_\mu(a) r_\nu(b)}{\sum_c r_\mu(c)} \mathbf{1}\{a \neq b\}, \end{aligned}$$

and d_g by breadth-first search in $\mathcal{C}_q(C_n)$ (Definition 2.7.1). Listings 1–4 concatenated form `figures.py`; Listing 5 is `verify.py`.

B.1. Shared core.

LISTING 1. Common definitions for Listings 2–4.

```

1 import itertools
2 import numpy as np
3 import matplotlib
4 matplotlib.use("Agg")
5 import matplotlib.pyplot as plt
6
7 plt.rcParams.update({
8     "font.family": "serif", "font.size": 9,
9     "mathtext.fontset": "cm",
10    "pdf.fonttype": 42, "ps.fonttype": 42,
11    "axes.linewidth": 0.6,
12    "xtick.direction": "in", "ytick.direction": "in",
13    "savefig.bbox": "tight", "savefig.pad_inches": 0.03})
14
15 def save(fig, name):

```

```

16     fig.savefig(name + ".pdf")
17     fig.savefig(name + ".png", dpi=400)
18     plt.close(fig)
19
20 def chain(q, n=4):
21     S = [c for c in itertools.product(range(1, q + 1), repeat=n)
22           if all(c[i] != c[(i + 1) % n] for i in range(n))]
23     idx = {c: i for i, c in enumerate(S)}
24     N = len(S)
25     P = np.zeros((N, N))
26     for c in S:
27         i = idx[c]
28         for v in range(n):
29             legal = [x for x in range(1, q + 1)
30                       if x not in {c[(v - 1) % n], c[(v + 1) % n]}]
31             for x in legal:
32                 d = list(c)
33                 d[v] = x
34                 P[i, idx[tuple(d)]] += 1.0 / (n * len(legal))
35     return S, P
36
37 def worst_tv(P, T):
38     N = len(P)
39     pi = np.full(N, 1.0 / N)
40     M = np.eye(N)
41     out = []
42     for _ in range(T + 1):
43         out.append(0.5 * np.abs(M - pi).sum(axis=1).max())
44         M = M @ P
45     return np.array(out)

```

B.2. Figure 5.

LISTING 2. Generation of Figure 5.

```

1 fig, ax = plt.subplots(figsize=(3.3, 2.2))
2 qs = np.arange(3, 13)
3 acl = 0.75 + 1 / (2 * (qs - 2))
4 apc = 0.75 + 1 / (qs - 2)
5 ax.axhline(1, ls="--", lw=0.7, color="black", zorder=1)
6 ax.plot([5, 5], [0.70, acl[2]], ":", lw=0.7, color="0.45", zorder=1)
7 ax.plot([7, 7], [0.70, apc[4]], ":", lw=0.7, color="0.45", zorder=1)
8 ax.plot(qs, acl, "o", ms=3.2, color="black", linestyle="none",
9         zorder=3, clip_on=False, label=r"$\alpha_{\mathrm{cl}}(q,4)$")
10 ax.plot(qs, apc, "o", ms=4.2, mfc="white", mec="black", mew=0.8,
11         linestyle="none", zorder=4, clip_on=False,
12         label=r"$\alpha_{\mathrm{pc}}(q,4)$")
13 ax.set_xlabel(r"$q$")
14 ax.set_xticks([3, 5, 7, 9, 11])
15 ax.set_ylim(0.70, 1.80)
16 ax.set_yticks([0.75, 1.00, 1.25, 1.50, 1.75])
17 ax.legend(frameon=False, fontsize=7, loc="upper right")
18 save(fig, "fig_alphas")

```

B.3. Figure 8.

LISTING 3. Generation of Figure 8.

```

1 fig, ax = plt.subplots(figsize=(4.6, 2.5))
2 markers = {3: "o", 4: "s", 5: "~"}
3 tmix = {}
4 ax.axhline(0.05, ls="--", lw=0.7, color="black", zorder=1)
5 for qq in (3, 4, 5):
6     d = worst_tv(chain(qq)[1], 60)
7     tmix[qq] = int(np.argmax(d <= 0.05))
8     ax.plot([tmix[qq] * 2, [0, 0.05]], ":", lw=0.7, color="0.45", zorder=1)
9     ax.plot(np.arange(61), d, markers[qq], ms=2.6, mfc="black", mec="black",
10            linestyle="none", zorder=3, clip_on=False,
11            label=r"$q={qq}$ ($t_{\mathrm{mix}}={tmix[qq]}$)")
12 ax.text(61.0, 0.05, r"$\varepsilon$", va="center", fontsize=8)
13 ax.set_xlabel(r"$t$")
14 ax.set_ylabel(r"$d(t)$")
15 ax.set_xlim(-1, 63)

```

```

16 ax.set_ylim(0, 1.02)
17 ax.legend(frameon=False, fontsize=7)
18 save(fig, "fig_decay")

```

B.4. Figure 2.

LISTING 4. Generation of Figure 2.

```

1 fig, ax = plt.subplots(figsize=(3.3, 2.3))
2 ns = np.arange(3, 11)
3 for qq, mk, mf in ((3, "o", "black"), (4, "s", "white")):
4     ax.semilogy(ns, (qq - 1.0) ** ns, "--", lw=0.7, color="0.45", zorder=1)
5     ax.semilogy(ns, (qq - 1.0) ** ns + (-1.0) ** ns * (qq - 1), mk, ms=3.4,
6                 mfc=mf, mec="black", mew=0.8, linestyle="none",
7                 zorder=3, clip_on=False,
8                 label=r"$|\Omega_{\{qq\}}(C_n)|$")
9 ax.text(10.2, 2.0 ** 10, r"$2^n$", fontsize=7, va="center")
10 ax.text(10.2, 3.0 ** 10, r"$3^n$", fontsize=7, va="center")
11 ax.set_xlabel(r"$n$")
12 ax.set_xticks([3, 5, 7, 9])
13 ax.set_xlim(2.6, 11.3)
14 ax.legend(frameon=False, fontsize=7, loc="upper left")
15 save(fig, "fig_growth")
16
17 print("wrote fig_{alphas,decay,growth}; t_mix:", tmix)

```

B.5. Verification.

LISTING 5. verify.py, certifying Table 1, Propositions 6.3.1 and 6.3.2, and Examples 5.1.3, 6.2.4, 6.2.5, and 6.2.6.

```

1 import itertools
2 from collections import deque
3 from fractions import Fraction
4 import numpy as np
5
6 def states(q, n):
7     return [c for c in itertools.product(range(q), repeat=n)
8             if all(c[i] != c[(i + 1) % n] for i in range(n))]
9
10 def legal(x, v, q, n):
11     return [c for c in range(q) if c not in {x[(v - 1) % n], x[(v + 1) % n]}]
12
13 def graph(q, n):
14     S = states(q, n)
15     idx = {c: i for i, c in enumerate(S)}
16     adj = [[] for _ in S]
17     for c in S:
18         for v in range(n):
19             for col in legal(c, v, q, n):
20                 if col != c[v]:
21                     d = list(c); d[v] = col
22                     adj[idx[c]].append(idx[tuple(d)])
23     return S, idx, adj
24
25 def all_dists(S, adj):
26     N = len(S); D = []
27     for s in range(N):
28         d = [-1] * N; d[s] = 0; Q = deque([s])
29         while Q:
30             u = Q.popleft()
31             for w in adj[u]:
32                 if d[w] < 0:
33                     d[w] = d[u] + 1; Q.append(w)
34     D.append(d)
35     return D
36
37 def tv_uniform(A, B):
38     return 1 - len(set(A) & set(B)) * Fraction(1, max(len(A), len(B)))
39
40 for q in (3, 4, 5, 6, 7):
41     S = states(q, 4); N = len(S)
42     idx = {c: i for i, c in enumerate(S)}

```

```

43 P = np.zeros((N, N))
44 for c in S:
45     i = idx[c]
46     for v in range(4):
47         L = legal(c, v, q, 4)
48         for col in L:
49             d = list(c); d[v] = col
50             P[i, idx[tuple(d)]] += 1.0 / (4 * len(L))
51 pi = np.full(N, 1.0 / N); M = np.eye(N); tmix = None
52 for t in range(200):
53     if tmix is None and 0.5 * np.abs(M - pi).sum(axis=1).max() <= 0.05:
54         tmix = t
55         M = M @ P
56 lam = float(np.sort(np.abs(np.linalg.eigvalsh(P)))[-2])
57 print(f"q={q} |0omega|={N} t_mix={tmix} lambda*={lam:.4f}")
58
59 for q in (3, 4, 5, 6, 7):
60     S = states(q, 4)
61     best = {k: Fraction(0) for k in (1, 2, 3, 4)}
62     for x in S:
63         for y in S:
64             S0 = [v for v in range(4) if x[v] != y[v]]
65             k = len(S0)
66             if k == 0:
67                 continue
68             E = Fraction(0)
69             for u in range(4):
70                 T = tv_uniform(legal(x, u, q, 4), legal(y, u, q, 4))
71                 E += Fraction(1, 4) * (k - (u in S0) + T)
72             best[k] = max(best[k], E / k)
73     ac1 = Fraction(3, 4) + Fraction(1, 2 * (q - 2))
74     print(f"q={q} classical maxima {dict(best)} alpha_c1={ac1}")
75
76 for q in (3, 4, 5, 6, 7):
77     S, idx, adj = graph(q, 4)
78     D = all_dists(S, adj)
79     best = Fraction(0)
80     for i, x in enumerate(S):
81         for j in adj[i]:
82             y = S[j]
83             E = Fraction(0)
84             for u in range(4):
85                 A, B = legal(x, u, q, 4), legal(y, u, q, 4)
86                 mA, mB = Fraction(1, len(A)), Fraction(1, len(B))
87                 mu = {c: (mA if c in A else Fraction(0)) for c in set(A) | set(B)}
88                 nu = {c: (mB if c in B else Fraction(0)) for c in set(A) | set(B)}
89                 rA = {c: mu[c] - min(mu[c], nu[c]) for c in mu}
90                 rB = {c: nu[c] - min(mu[c], nu[c]) for c in nu}
91                 T = sum(rA.values())
92                 for c in mu:
93                     w = min(mu[c], nu[c])
94                     if w:
95                         X = list(x); X[u] = c; Y = list(y); Y[u] = c
96                         E += Fraction(1, 4) * w * D[idx[tuple(X)]] [idx[tuple(Y)]]
97             if T:
98                 for ca, wa in rA.items():
99                     if not wa: continue
100                     for cb, wb in rB.items():
101                         if not wb: continue
102                         X = list(x); X[u] = ca; Y = list(y); Y[u] = cb
103                         E += Fraction(1, 4) * wa * wb / T * D[idx[tuple(X)]] [idx[tuple(Y)]]
104             best = max(best, E)
105     apc = Fraction(3, 4) + Fraction(1, q - 2)
106     print(f"q={q} path maximum {best} alpha_pc={apc}")
107
108 for q in (3, 4, 5, 6, 7):
109     S, idx, adj = graph(q, 4)
110     D = all_dists(S, adj)
111     dG = max(max(r) for r in D)
112     swaps = sum(1 for i, x in enumerate(S) for j, y in enumerate(S)
113                 if sum(a != b for a, b in zip(x, y)) == 2 and D[i][j] == 3)
114     gap = max(D[i][j] - sum(a != b for a, b in zip(S[i], S[j]))
115              for i in range(len(S)) for j in range(len(S)))
116     print(f"q={q} diam_G={dG} max(dG-dH)={gap} swap pairs={swaps}")
117
118 for (n, q) in ((3,4),(3,5),(4,3),(4,4),(4,5),(5,4),(5,5),(6,4),(7,4)):
119     S, idx, adj = graph(q, n)
120     dG = max(max(r) for r in all_dists(S, adj))
121     print(f"n={n} q={q} diam_G={dG} floor(3n/2)={3*n//2}")

```



```

122
123 def replay(z, moves, y, q, n):
124     z = list(z)
125     for v, c in moves:
126         assert c != z[v] and c in legal(z, v, q, n)
127         z[v] = c
128     assert tuple(z) == y
129     return len(moves)
130
131 print("C5 schedule:", replay((0,1,0,1,2),
132     [(1,4),(3,4),(0,1),(2,1),(4,3),(1,0),(3,0)], (1,0,1,0,3), 5, 5), "moves")
133 print("C6 schedule:", replay((0,1,0,1,0,1),
134     [(0,2),(2,2),(4,2),(1,0),(3,0),(5,0),(0,1),(2,1),(4,1)], (1,0,1,0,1,0), 5, 6), "moves")

```

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